Indian Council of Medical Research

ICMR AMRIT

Operator manual

One system, two programs. A laboratory runs the desktop application; a country runs the central server. Between them travels nothing but aggregate numbers. Every screenshot in this manual is the real software, captured from a running installation.

Deployment	Contract version	Issued
India	2.0	2026-08-14

Antimicrobial Resistance Monitoring, Intelligence and Tracking

This manual covers both halves of the system and the reasoning behind them:

- **Part I — the desktop application**, which a laboratory runs on its own computer, offline.
- **Part II — the central web portal**, which a programme runs for a country or a network.
- **Part III — methods and rationale**, which states what each analytical choice is, what the alternatives were, and why this one was taken. Part III is written to be quotable in a paper: every method names the module that implements it.

Every screenshot is captured from a running build — the desktop screens by the application's own `AMRIT_CAPTURE_DIR` mode against a freshly seeded demonstration database, the portal screens by `tools/capture_portal.mjs` signed in as each role — and every figure is generated by `tools/build_manual_figures.py`. Neither is drawn by hand, because a picture that is edited separately from the system drifts away from it and nobody notices until a reader follows it and is wrong.

Part I — Desktop application

1. What it is for, and what it refuses to do

A microbiology laboratory produces isolate records: a patient, a specimen, an organism, and a panel of antimicrobial susceptibility results. AMRIT's desktop application is where those records live. It is a local-first application — Electron over SQLite — and it is fully functional with no network at all. A laboratory can install it, type records for a year, analyse them, print an antibiogram and export a WHONET file without ever connecting to anything.

It refuses three things by construction:

- It never uploads a patient row. What leaves this application over the network is a count or a rate (§27, §28).
- It never interprets a susceptibility result against a breakpoint set nobody activated (§23).

- It never invents data. A value it cannot read, map or interpret is reported and left blank, not guessed.

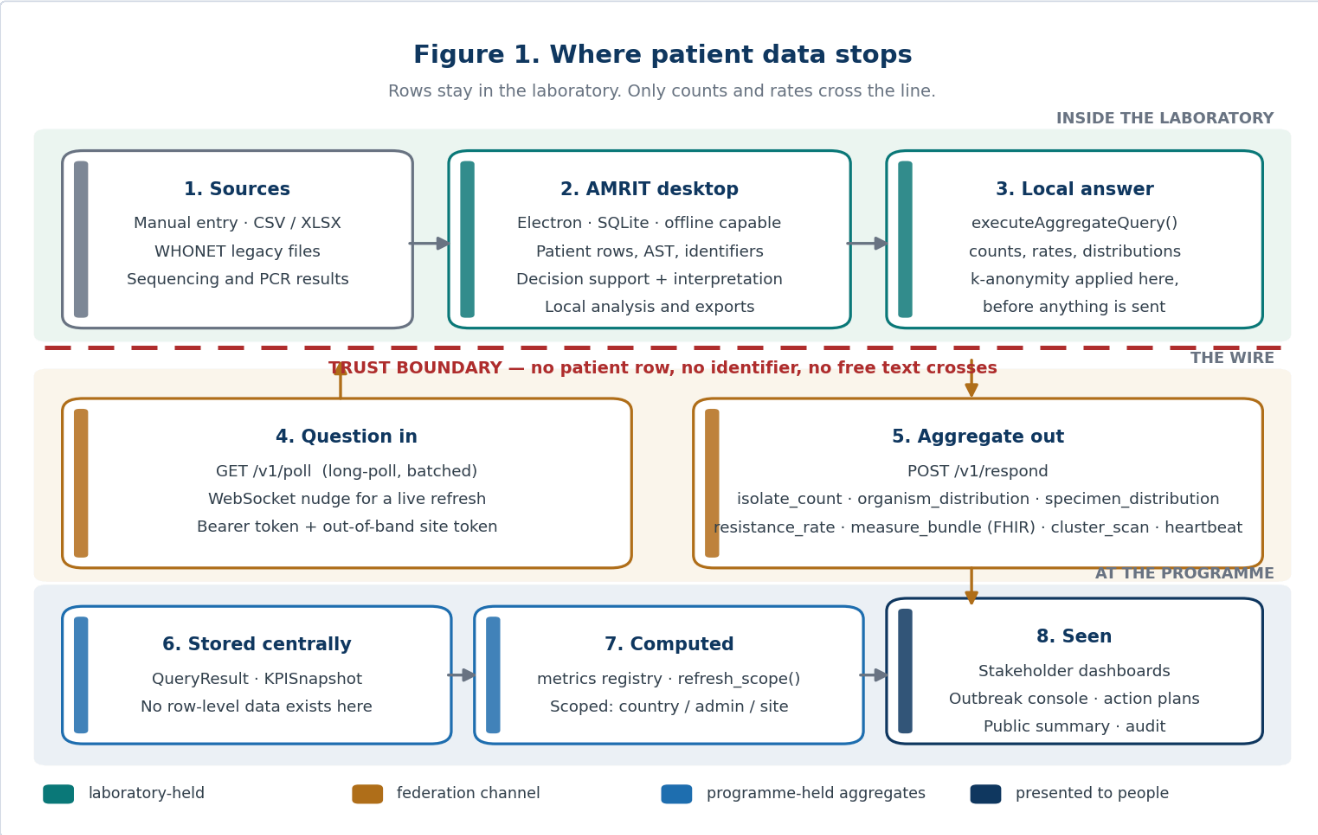
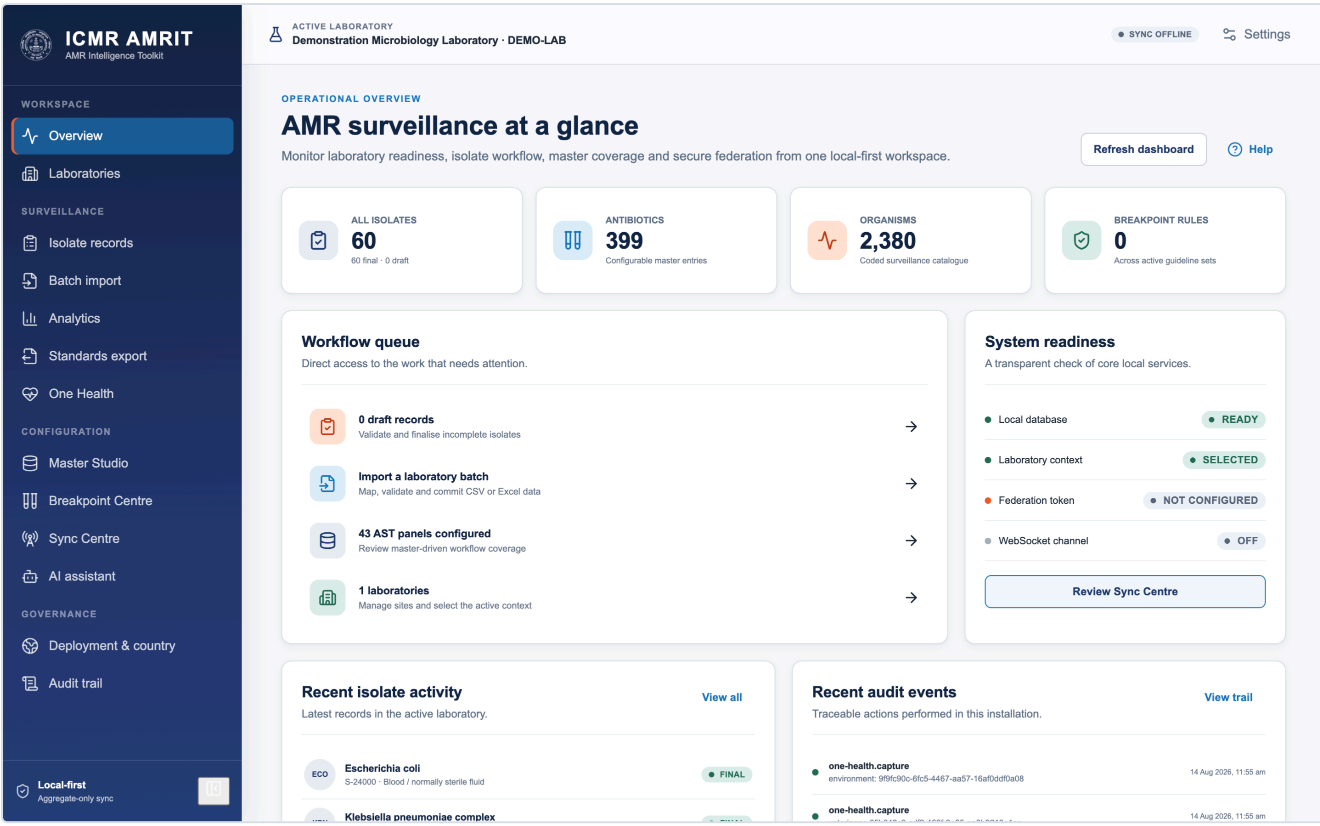


Figure 1 — where patient data stops.



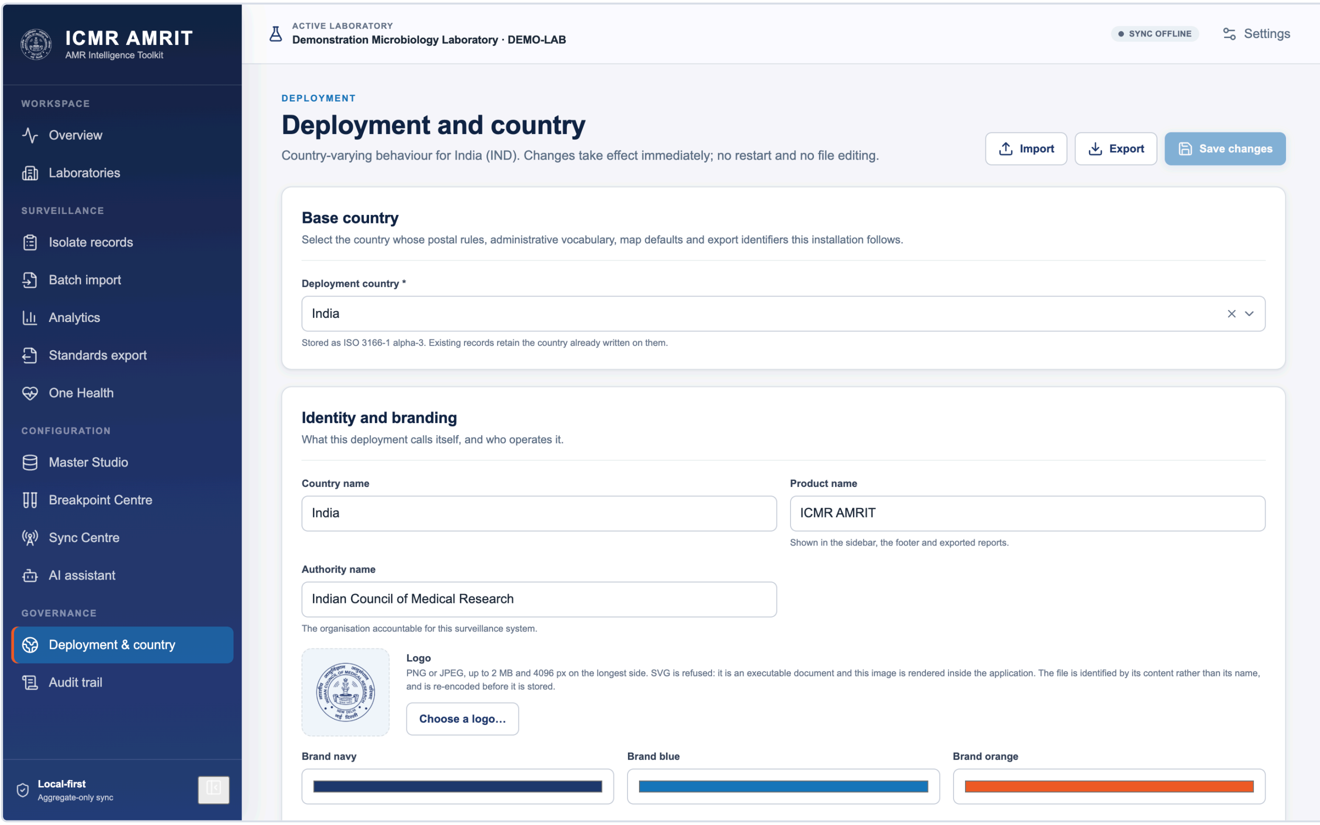
2. Install and first run

Install the build for your platform (`.dmg`, `.exe`, `.AppImage`). On first run the application creates its database under the operating system's application-data directory and seeds the packaged catalogue: WHONET organisms, antimicrobials, specimen types, expert rules and panels, hash-pinned so a tampered catalogue is refused rather than loaded.

Nothing else is created. There is no default account, no default laboratory and no sample data — a deployment holding real results must never inherit invented ones.

3. Configure the deployment

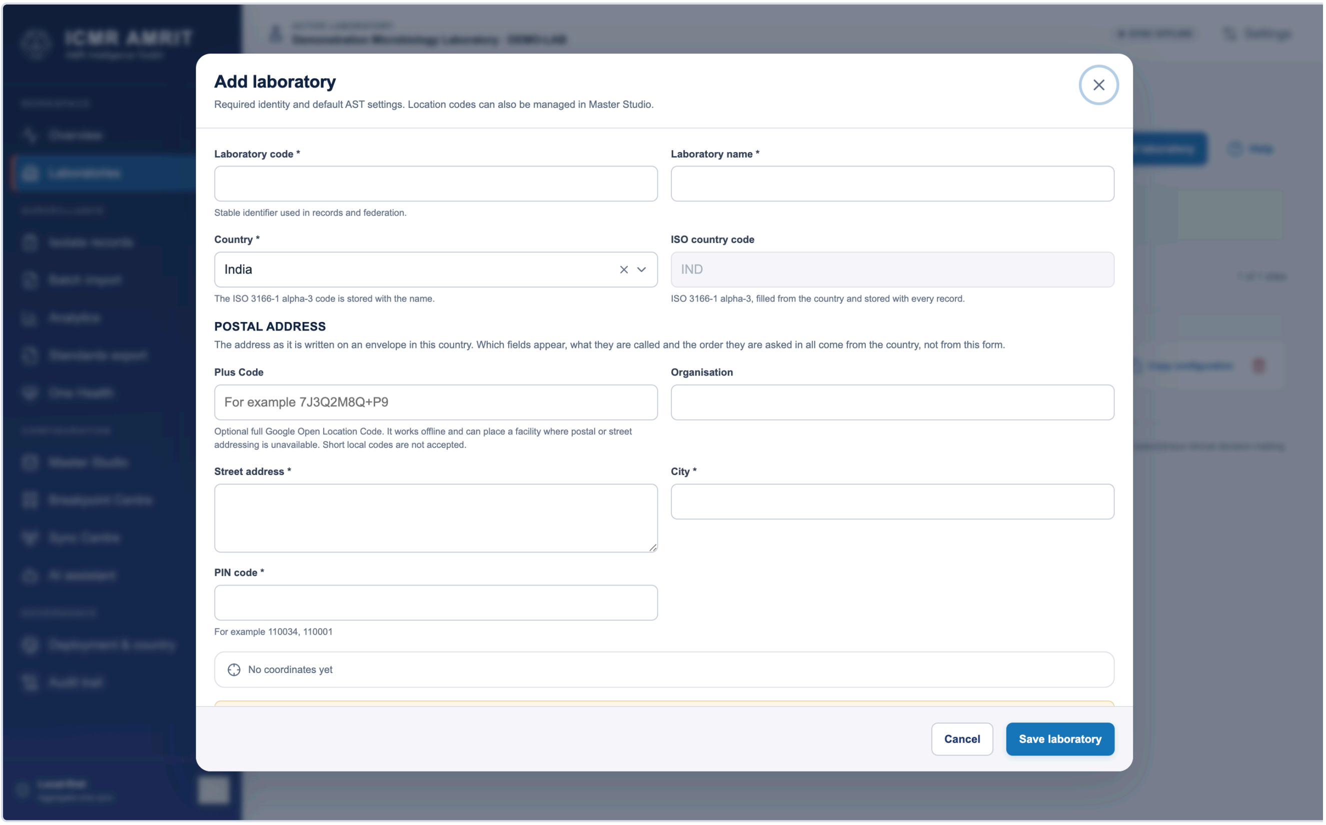
Deployment → country profile. The profile decides the country, its administrative tiers (what they are called and how many there are), the default guideline (CLSI or EUCAST), identifier namespaces, branding and the retention period. A profile is data, not code: the same binary serves any country, and `python3 tools/generate_country_reference.mjs` scaffolds a new one.



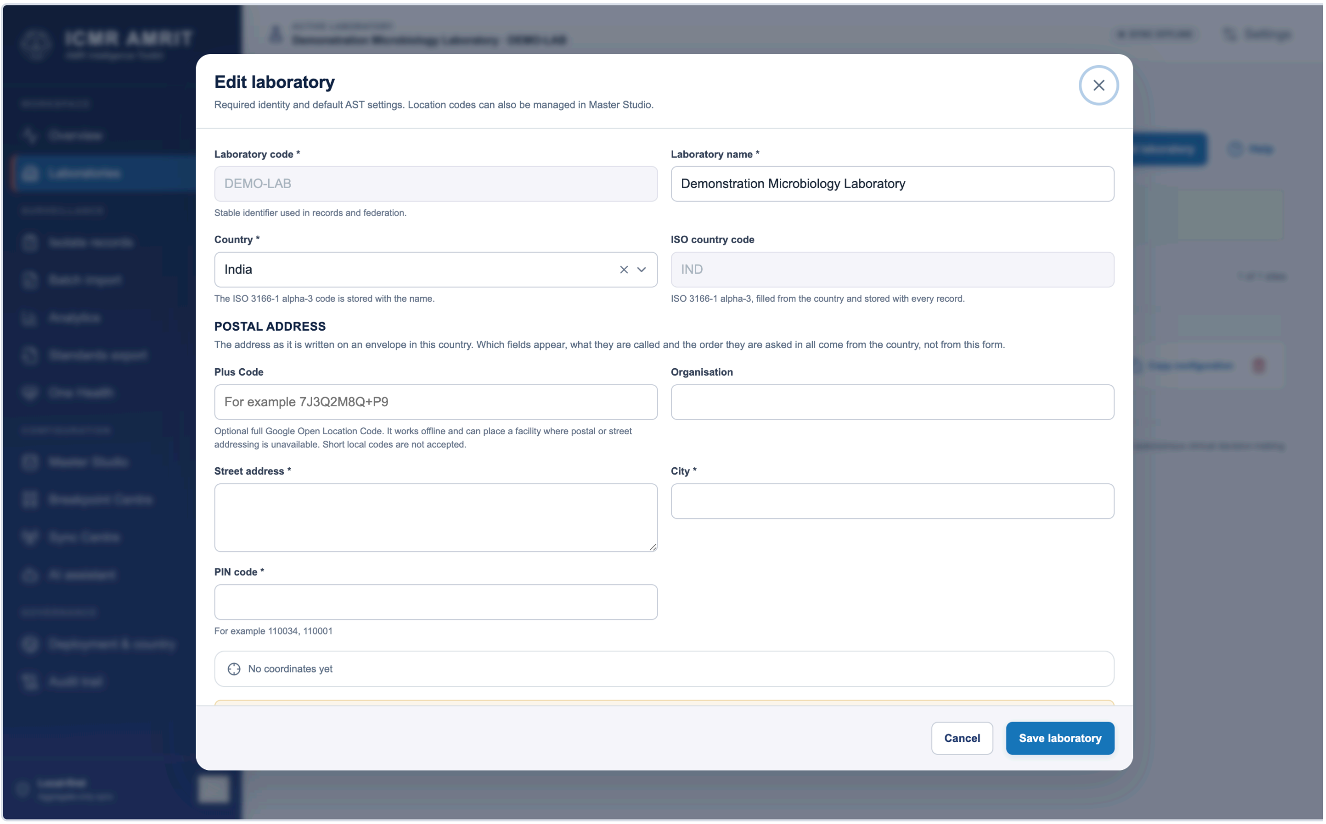
Two changes are irreversible and say so before they are saved: changing the identifier namespace (existing identifiers keep the old one) and changing the country (geography and overrides from the previous country are discarded).

4. Register the laboratory and place it on the map

Laboratories → New laboratory. A laboratory needs a code, a name and an address in the country's own address shape — the form is generated from the profile, so an Indian deployment asks for state and district and a Testland deployment asks for governorate and district.



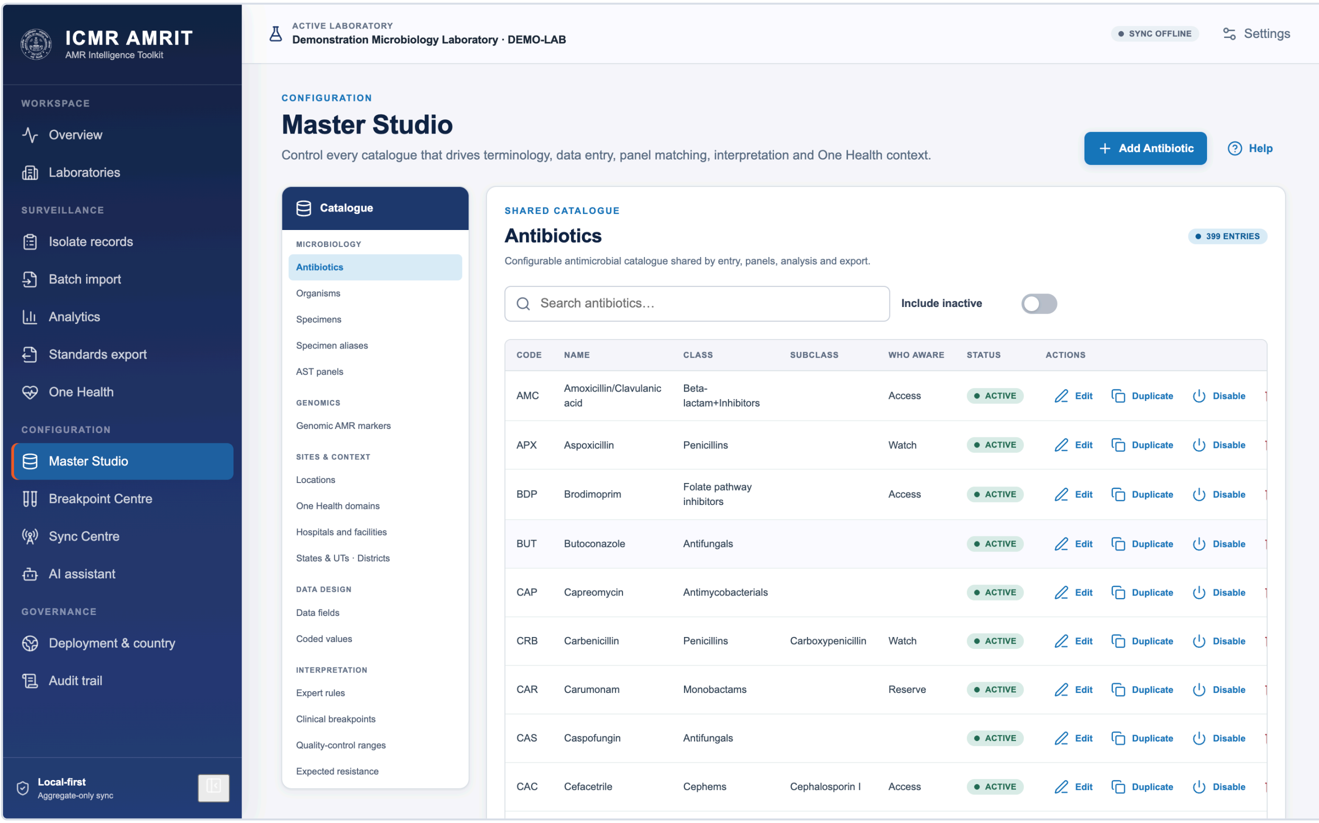
The address resolves to an administrative unit from the bundled geo pack. That unit — not a typed place name — is what every scope filter, dashboard roll-up and map placement uses afterwards.



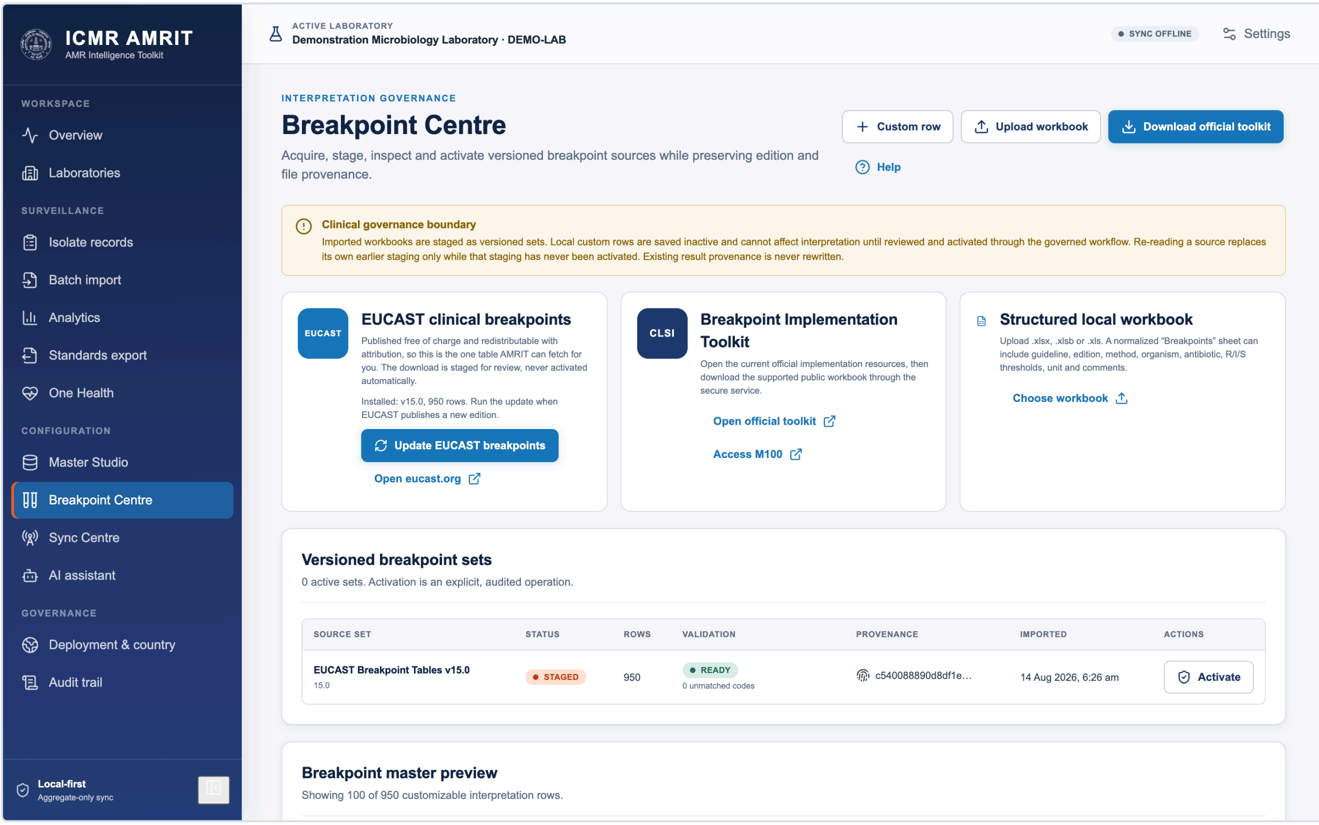
Site coordinates are optional and consent-gated (Sync → Optional location sharing). With consent given, Use this computer's location reads the operating system's location service; the coordinates can always be typed instead, and the source (device or hand-entered) travels with the heartbeat, because how a coordinate was obtained is part of what it means.

5. Catalogues and breakpoints

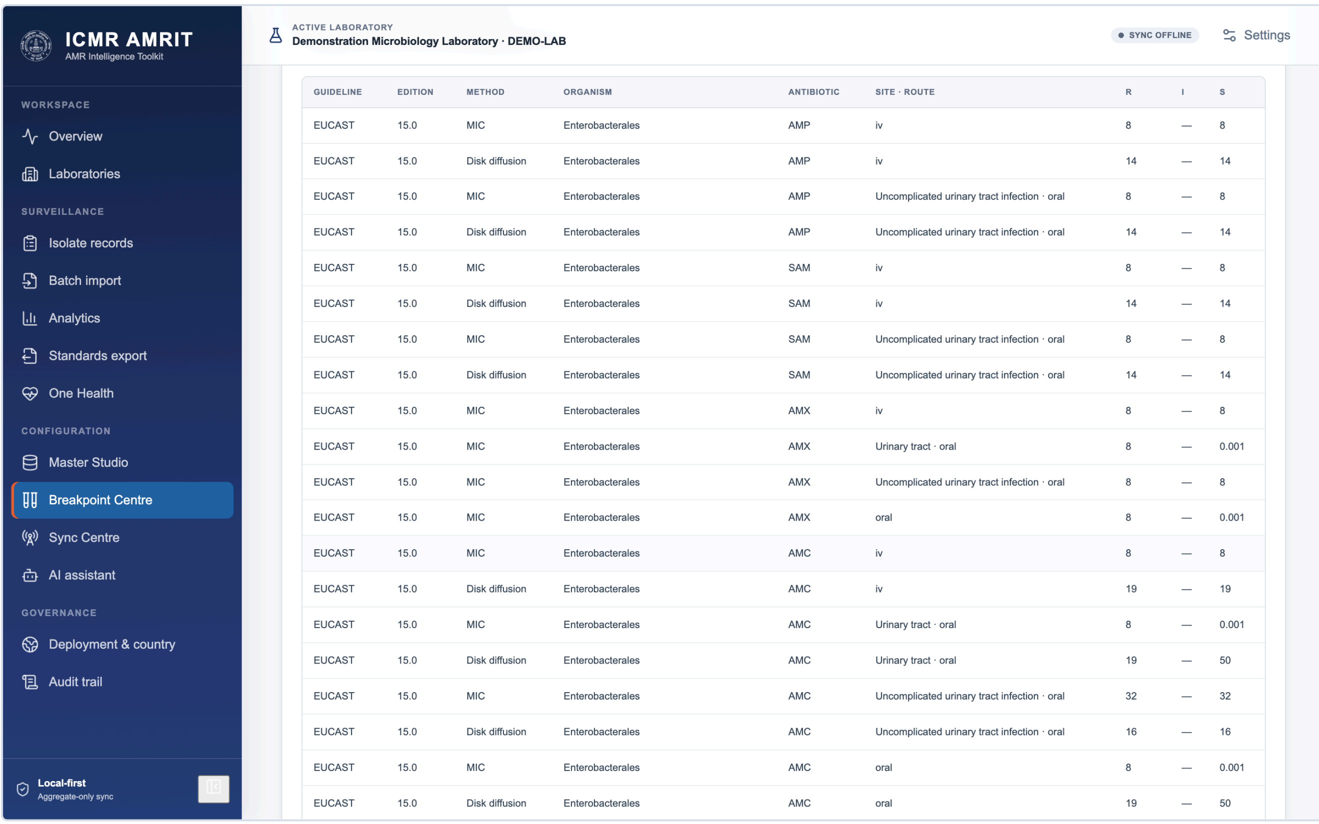
Master Studio holds the reference data: organisms, antimicrobials, specimens, panels, expected-resistance rules, expert rules, and local additions. Packaged rows are read-only; local rows are marked custom and can be deactivated but never silently overwrite the catalogue.



Breakpoint Centre is where guidelines arrive. EUCAST publishes its tables free of charge and permits redistribution, so the application fetches and stages the current edition with one button. CLSI's M100 is a paid standard: the application links to it and imports a workbook you supply. Either way the set is *staged inactive* and must be activated deliberately (§23).



The screenshot above is a live fetch: EUCAST v15.0, 950 rows staged, `0 unmatched codes`, `READY`, and an **Activate** button that has not been pressed. Until it is, nothing in this set interprets anything.



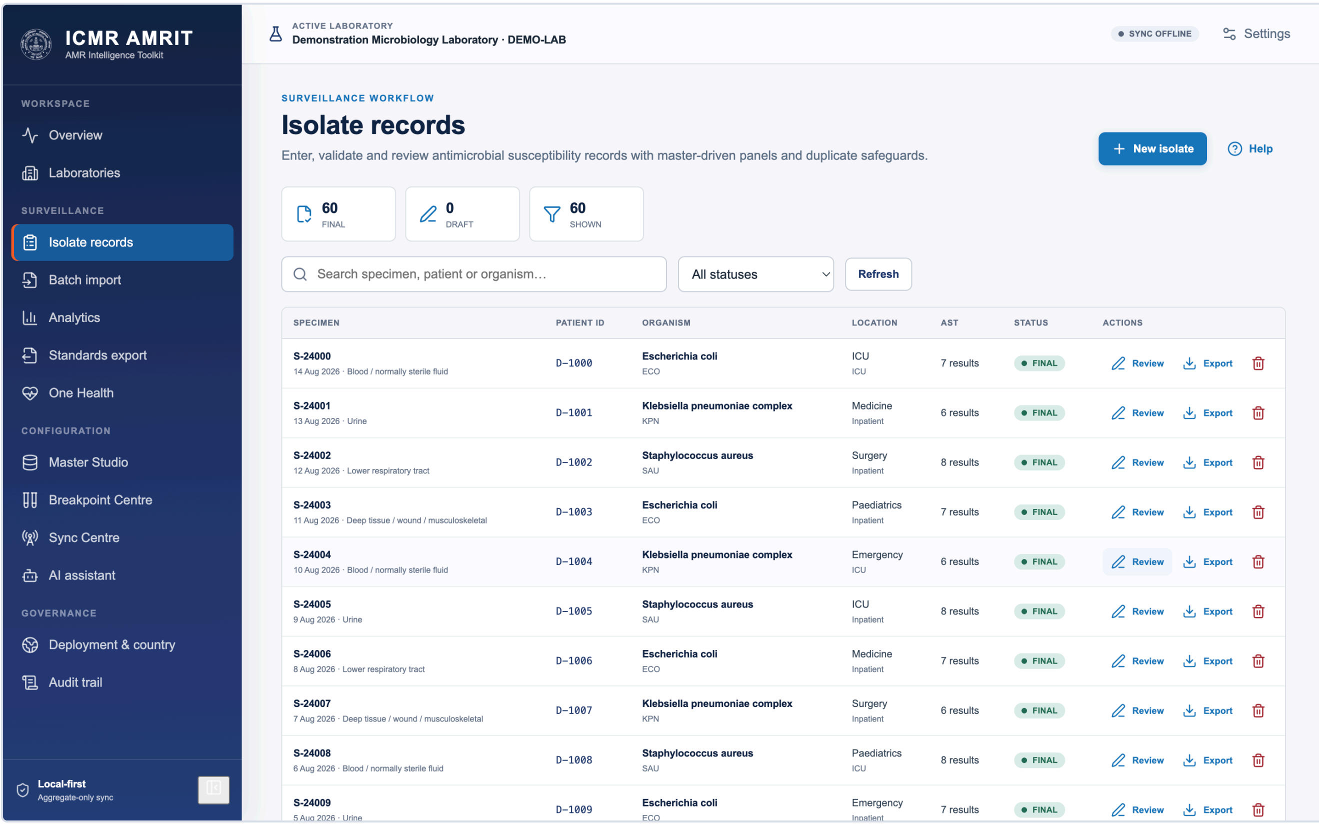
GUIDELINE	EDITION	METHOD	ORGANISM	ANTIBIOTIC	SITE · ROUTE	R	I	S
EUCAST	15.0	MIC	Enterobacterales	AMP	iv	8	—	8
EUCAST	15.0	Disk diffusion	Enterobacterales	AMP	iv	14	—	14
EUCAST	15.0	MIC	Enterobacterales	AMP	Uncomplicated urinary tract infection · oral	8	—	8
EUCAST	15.0	Disk diffusion	Enterobacterales	AMP	Uncomplicated urinary tract infection · oral	14	—	14
EUCAST	15.0	MIC	Enterobacterales	SAM	iv	8	—	8
EUCAST	15.0	Disk diffusion	Enterobacterales	SAM	iv	14	—	14
EUCAST	15.0	MIC	Enterobacterales	SAM	Uncomplicated urinary tract infection · oral	8	—	8
EUCAST	15.0	Disk diffusion	Enterobacterales	SAM	Uncomplicated urinary tract infection · oral	14	—	14
EUCAST	15.0	MIC	Enterobacterales	AMX	iv	8	—	8
EUCAST	15.0	MIC	Enterobacterales	AMX	Urinary tract · oral	8	—	0.001
EUCAST	15.0	MIC	Enterobacterales	AMX	Uncomplicated urinary tract infection · oral	8	—	8
EUCAST	15.0	MIC	Enterobacterales	AMX	oral	8	—	0.001
EUCAST	15.0	MIC	Enterobacterales	AMC	iv	8	—	8
EUCAST	15.0	Disk diffusion	Enterobacterales	AMC	iv	19	—	19
EUCAST	15.0	MIC	Enterobacterales	AMC	Urinary tract · oral	8	—	0.001
EUCAST	15.0	Disk diffusion	Enterobacterales	AMC	Urinary tract · oral	19	—	50
EUCAST	15.0	MIC	Enterobacterales	AMC	Uncomplicated urinary tract infection · oral	32	—	32
EUCAST	15.0	Disk diffusion	Enterobacterales	AMC	Uncomplicated urinary tract infection · oral	16	—	16
EUCAST	15.0	MIC	Enterobacterales	AMC	oral	8	—	0.001
EUCAST	15.0	Disk diffusion	Enterobacterales	AMC	oral	19	—	50

The preview lists what was staged — organism scope, agent, method, site and route, and the R/I/S thresholds — so a reviewer checks rows rather than trusting a row count.

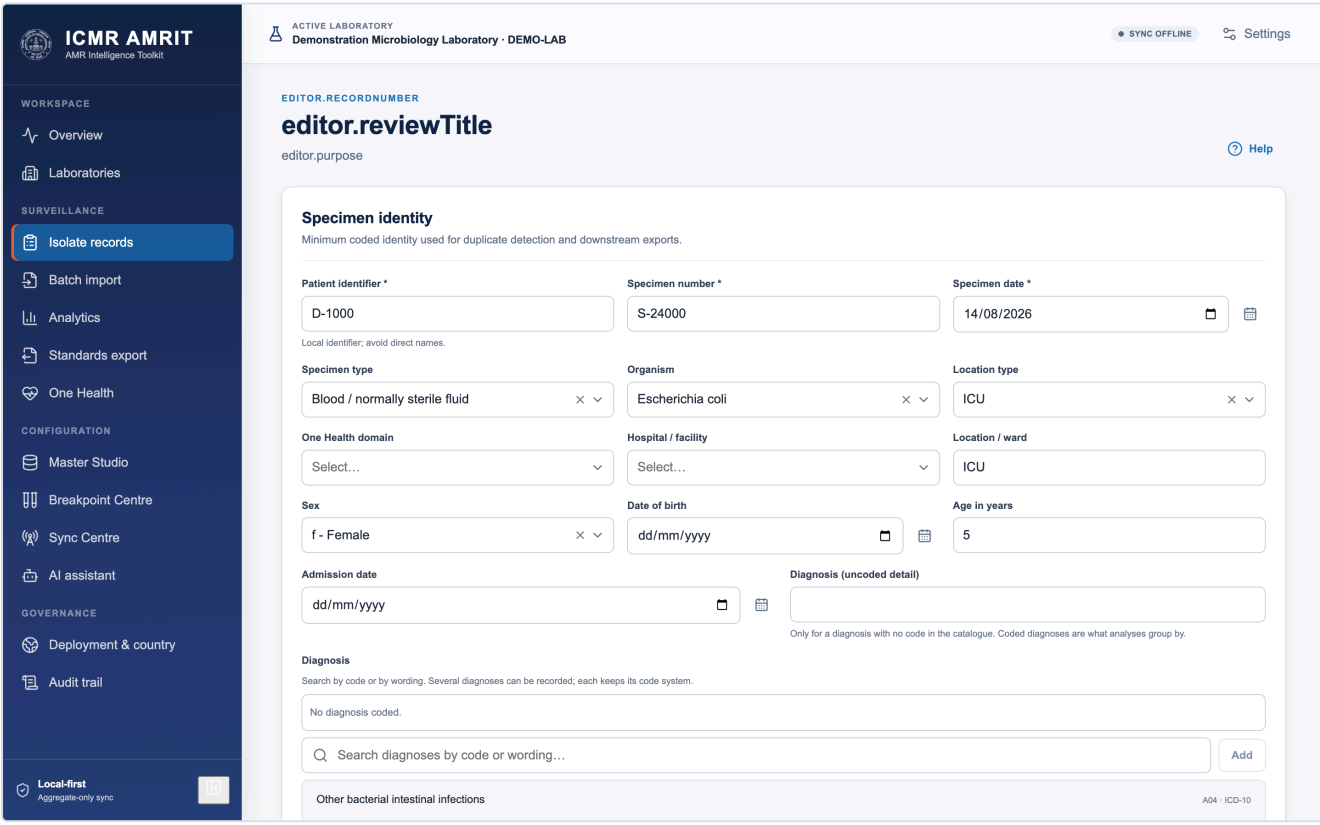
6. Enter data

Three routes, all landing in the same validated shape:

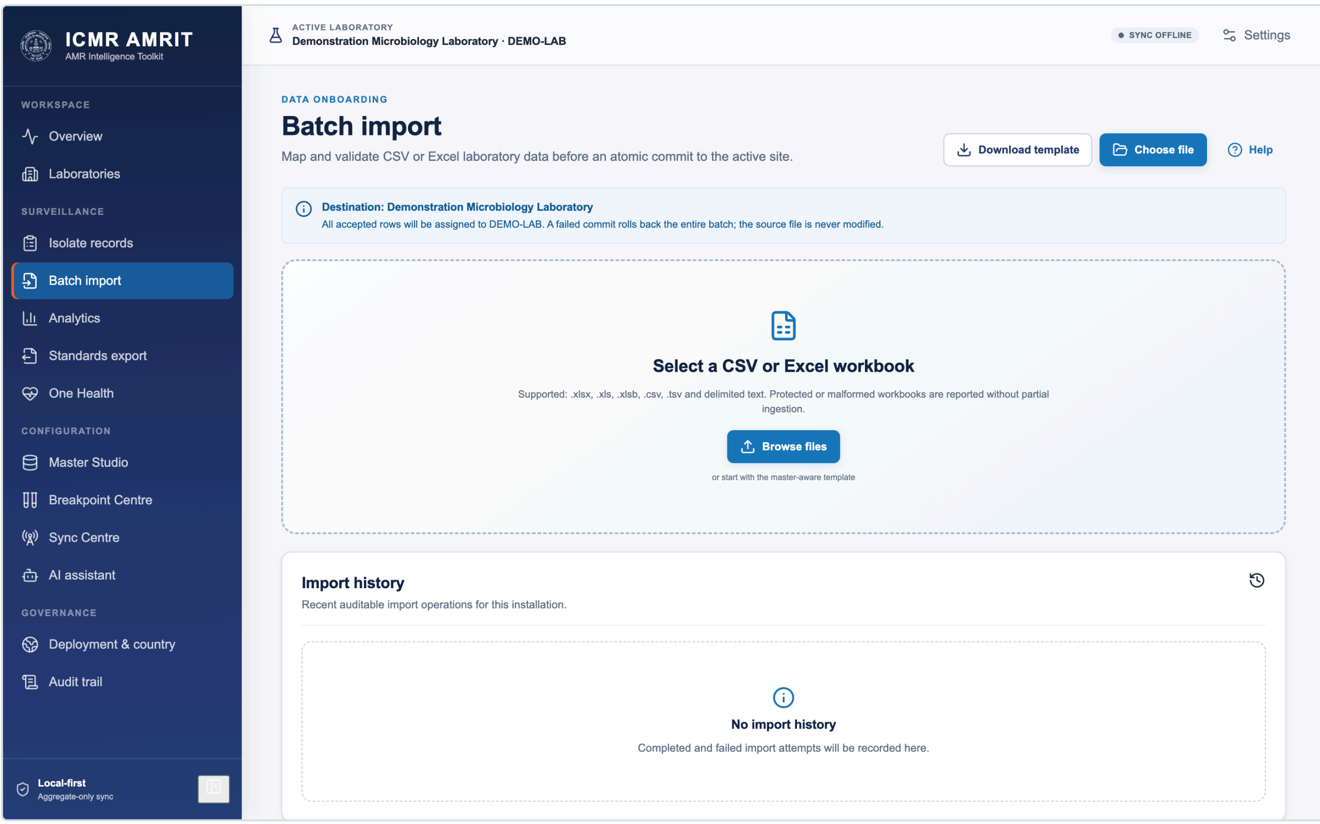
Type a record. Identity first (patient identifier, specimen number and date, specimen type, organism), then residence, then AST. The antimicrobial panel offered is the one matching the organism and specimen, from the panel catalogue.



The editor is where identity, specimen, organism and the antimicrobial panel are entered, and where decision support answers as you type:



Import a batch. Map columns once, save the mapping as a profile, and reuse it. The preview shows what will be imported, what will land as a draft and what is refused, per row and with a reason, before anything is written.



WHONET legacy. Existing WHONET data files import directly, including their code sets.

7. Quality control and decision support

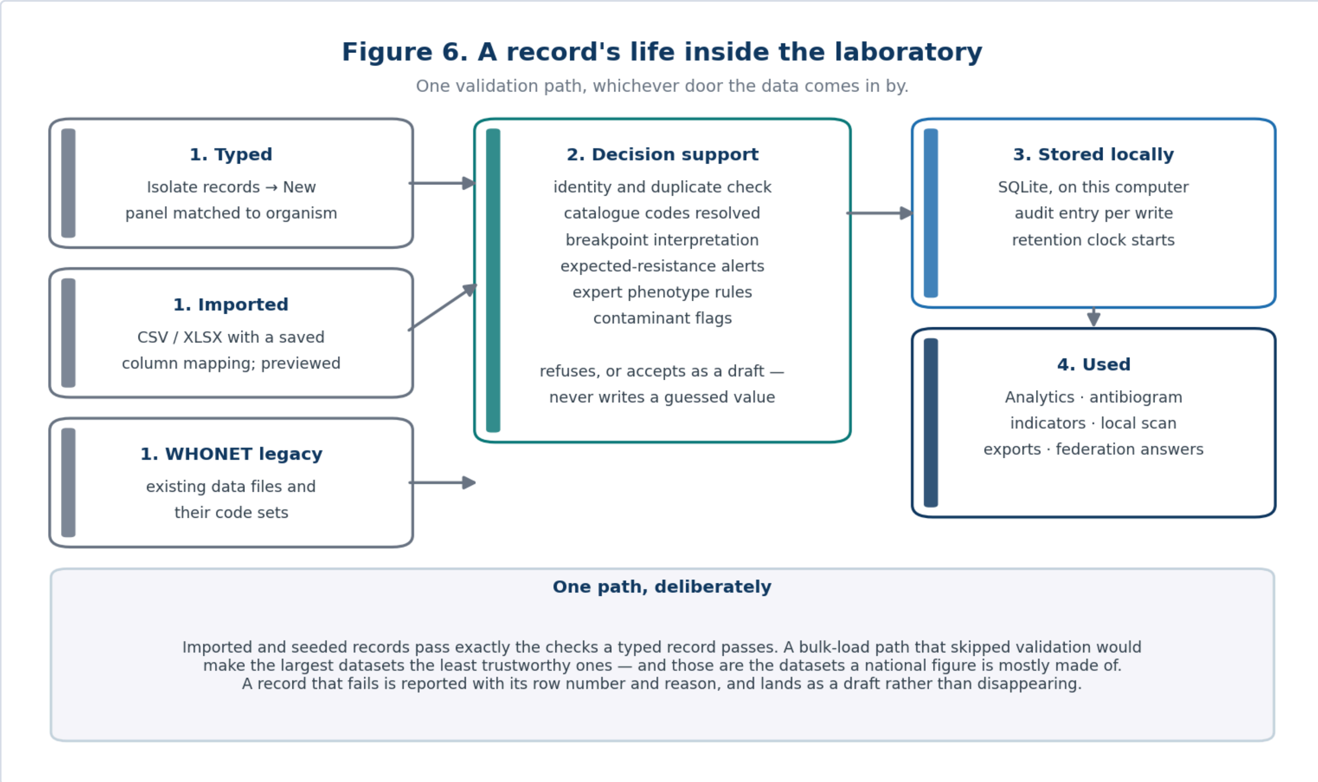


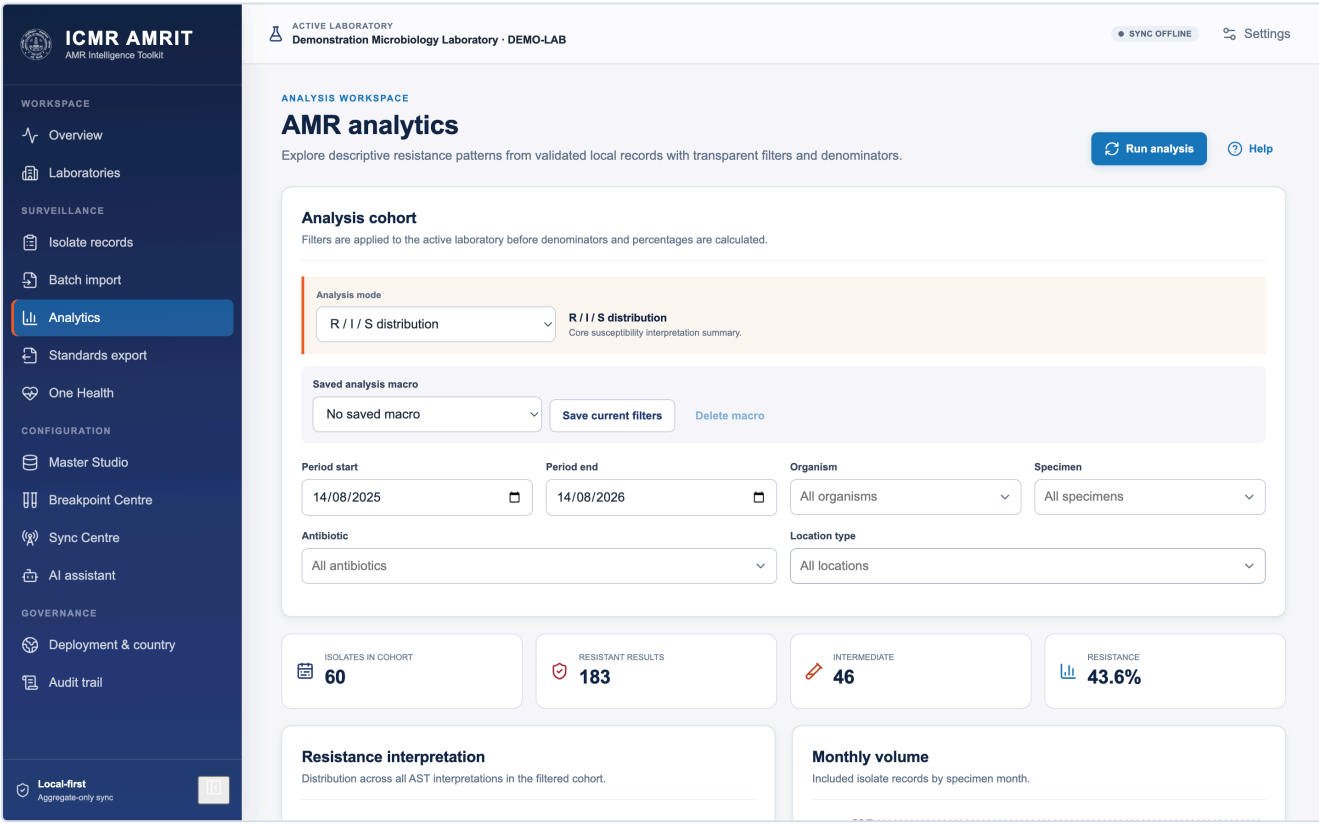
Figure 6 — one validation path, whichever door the data comes in by.

Every write passes the same decision-support layer, whether typed, imported or seeded:

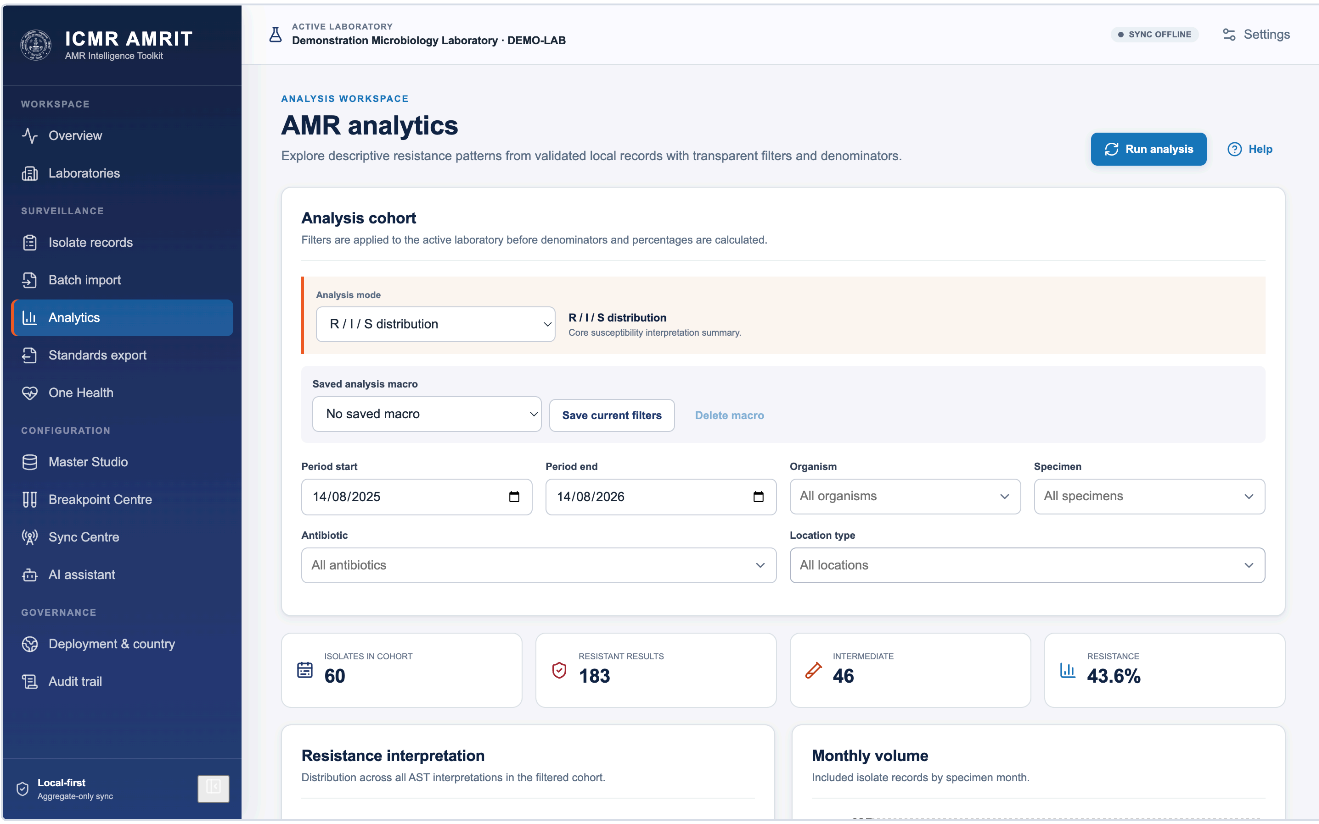
- Interpretation. A measurement with no explicit S/I/R is interpreted from the active breakpoint set — and left blank with a stated reason when the set has no matching row, when two equally specific rows disagree, or when the record does not state something the breakpoint depends on (§23).
- Expected resistance. An organism reported susceptible to an agent it is intrinsically resistant to raises an alert rather than being corrected silently.
- Expert rules. Configurable phenotype rules (ESBL, carbapenemase, MRSA, inducible clindamycin) add comments and alerts.
- Contaminant and duplicate checks. Likely skin flora in a sterile-site culture is flagged; an identical patient/specimen identity is refused as a duplicate.

8. Analyse

Analytics runs deterministic, reproducible analyses over the local database: resistance rates by organism and agent, antibiograms, specimen and organism distributions, priority indicators, time trends, and a local outbreak scan.



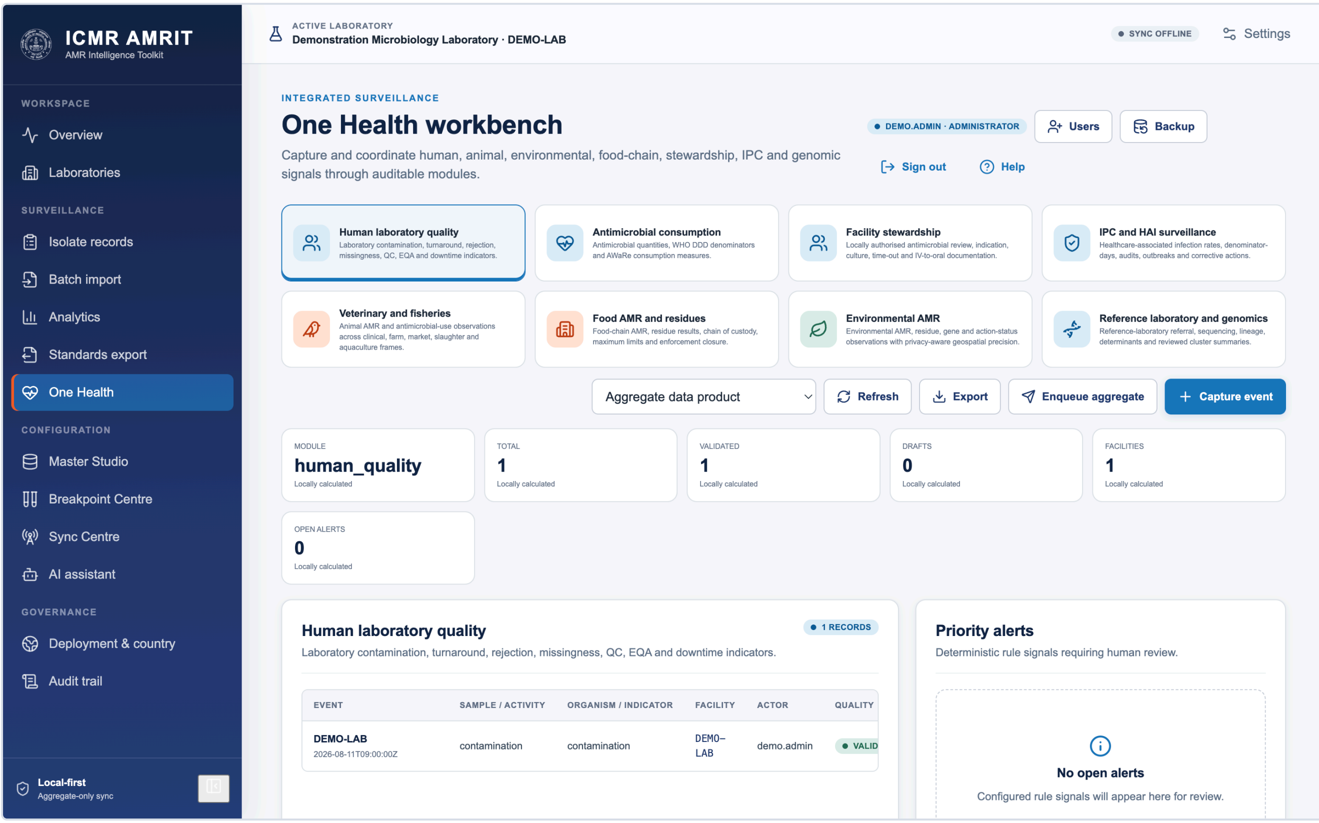
With an analysis run, the same screen carries the antibiogram, the priority indicators and the resistance table computed over the current filter set:



Two things are always explicit on screen because they change every number: the **deduplication mode** (§24) and the **date window**. A saved macro stores a whole filter set so the same analysis can be re-run next month and compared honestly.

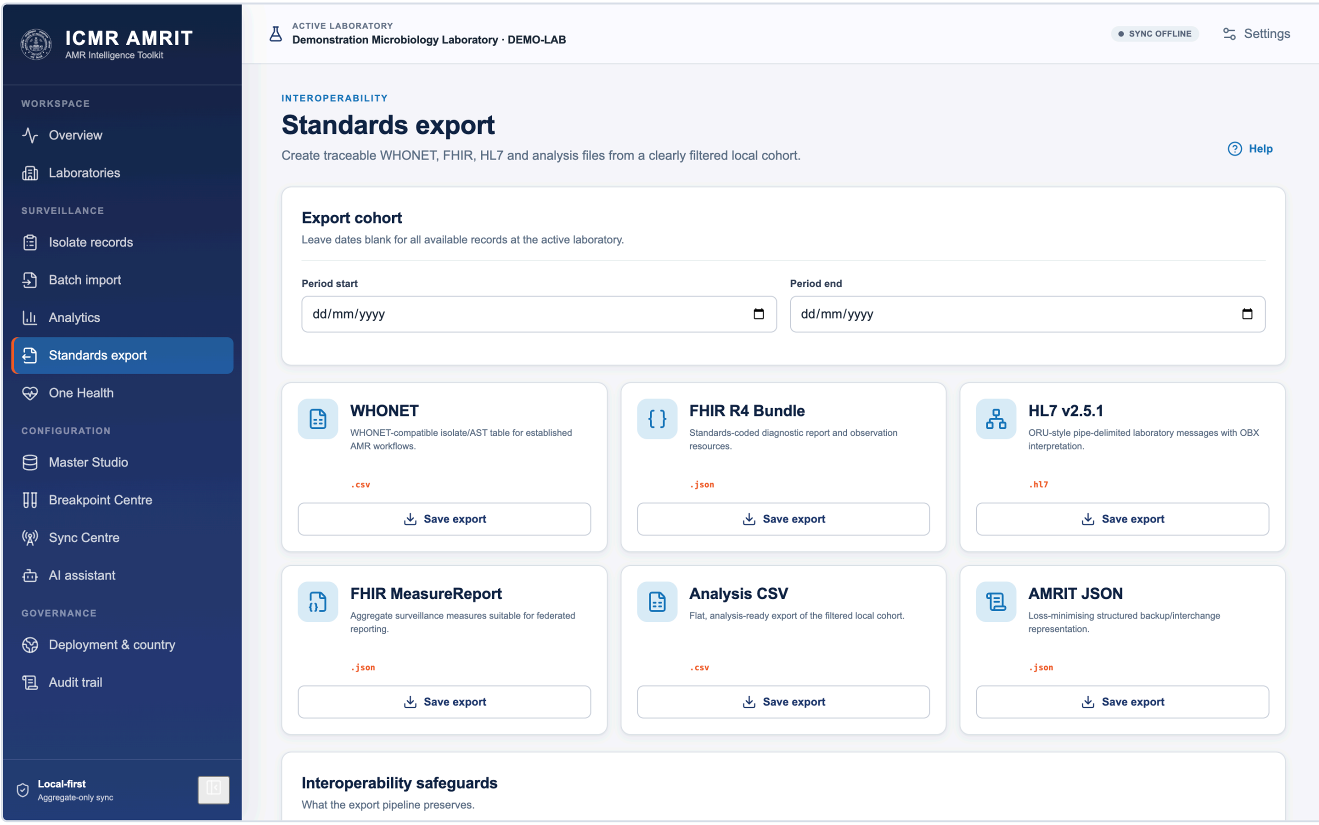
9. One Health capture

AMR is not only a human-health problem. The **One Health** workbench captures veterinary, environmental, food-chain, antimicrobial-consumption and laboratory-quality events against the same facility registry, with its own operator accounts and its own governance.



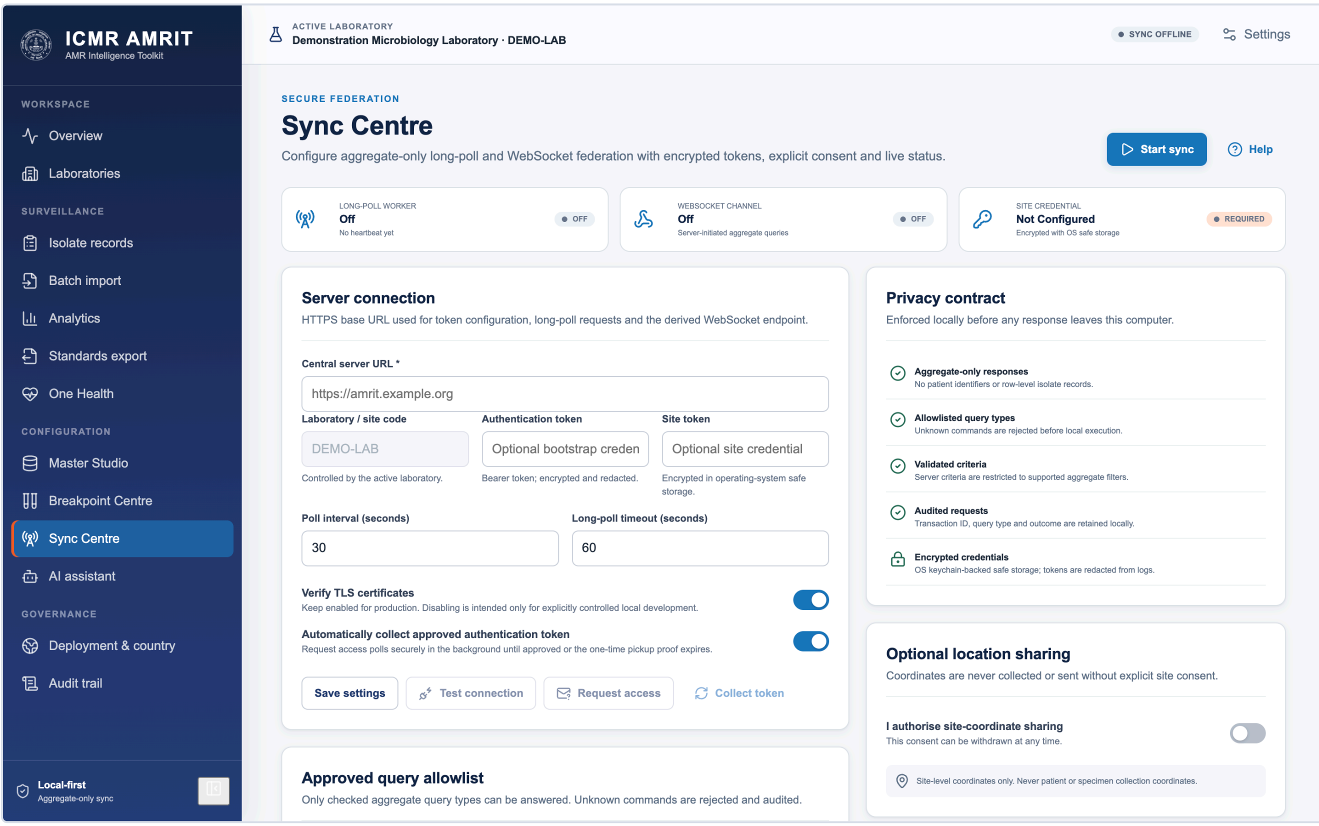
10. Export

Format	What it is for
WHONET	Exchange with the WHONET/GLASS ecosystem
FHIR (Observation / MeasureReport)	Interoperable clinical and aggregate reporting
HL7 v2	Feeding a hospital information system
Measure bundle	Aggregate indicator reporting to a programme
CSV / JSON	Analysis in R, Python, Excel



An export is a deliberate act by the operator, and it may contain patient-level data — that is the difference between an export and federation (§28), and the audit log records both.

11. Join a central server (federation)



1. **Request access.** The desktop registers its lab code with the server and receives a one-time *pickup token*. Nothing is granted yet.
2. **An administrator approves** it in the portal, which creates the site and mints a *site token* delivered out of band.
3. **Collect the token.** The desktop redeems its pickup token for a bearer token it mints and stores in the operating system's credential vault. The server keeps only a hash.
4. **Allow-list the query types** this laboratory will answer. A query type not on the list is refused, logged and answered with an error.
5. **Start sync.** A long-poll worker asks for questions; a WebSocket connection lets the server nudge for an immediate answer. A four-hourly heartbeat reports presence.

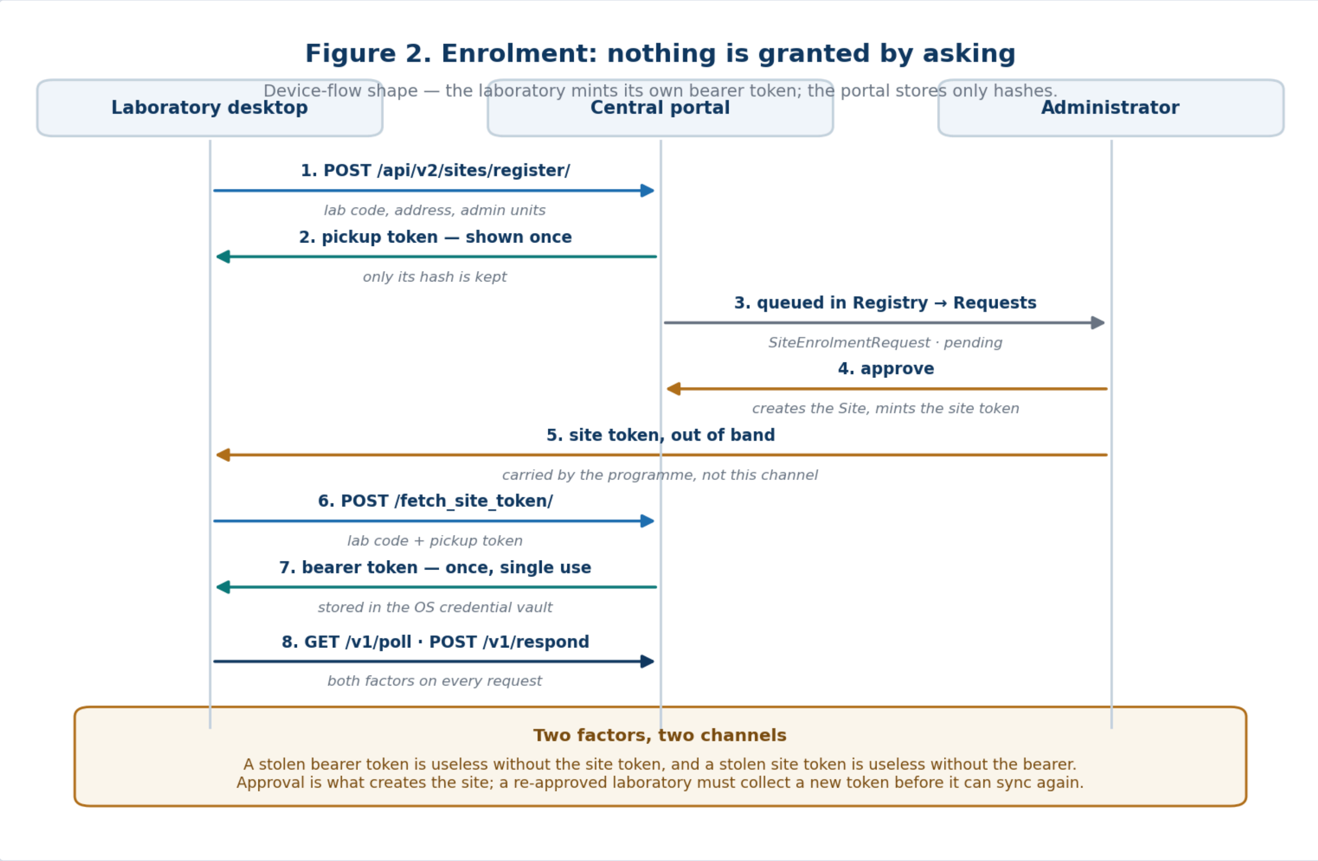
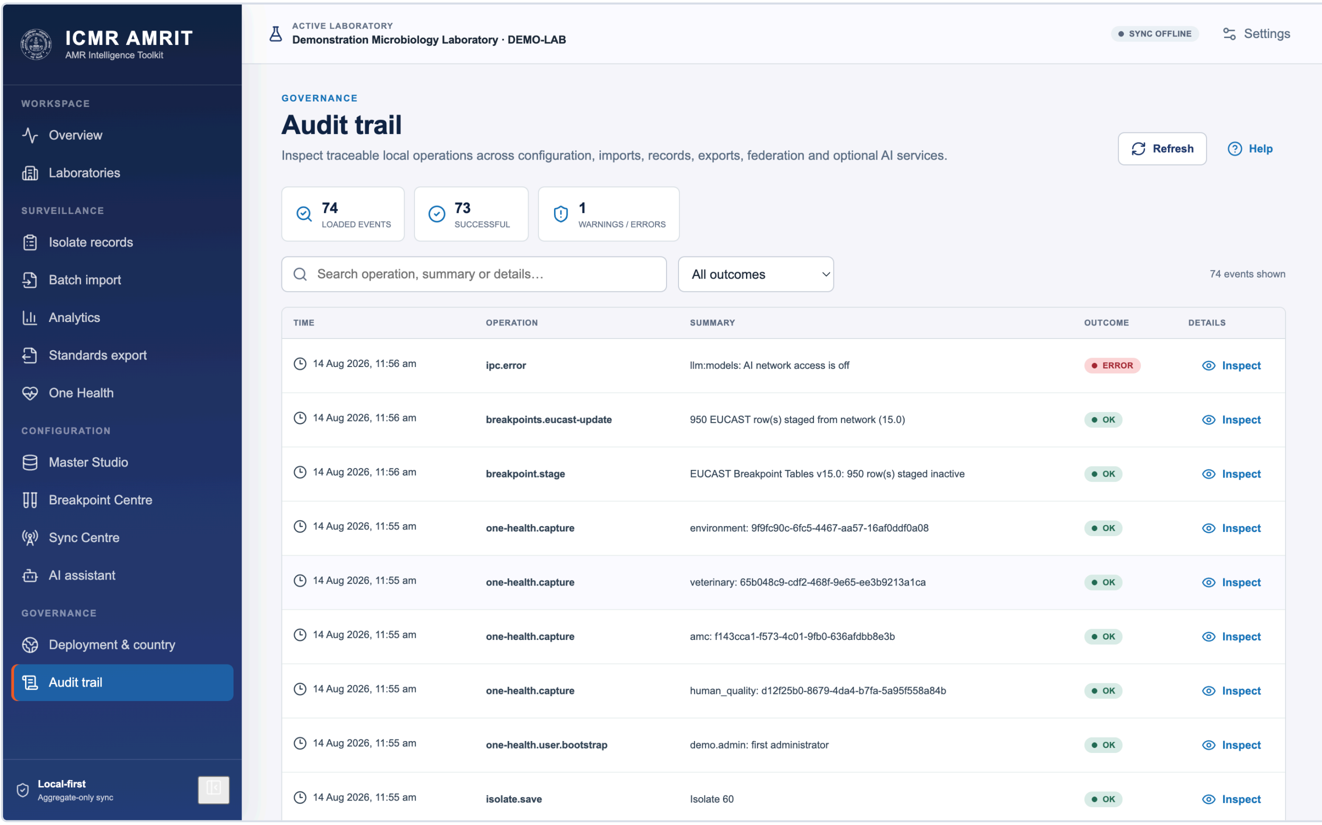


Figure 2 — enrolment and the two-token credential exchange.

12. Privacy, retention, audit

- **Retention.** The profile sets a retention period; expiry removes row-level data on a schedule, with a dry run that reports what would go first.
- **Erasure.** A named subject's rows can be erased on request, with the erasure itself audited.
- **Audit.** Every consequential act — saving a record, activating a breakpoint set, exporting a file, starting sync, answering a query — is written to an append-only local log.


ICMR AMRIT
AMR Intelligence Toolkit
WORKSPACE
Overview
Laboratories
SURVEILLANCE
Isolate records
Batch import
Analytics
Standards export
One Health
CONFIGURATION
Master Studio
Breakpoint Centre
Sync Centre
AI assistant
GOVERNANCE
Deployment & country
Audit trail
Local-first
Aggregate-only sync
ACTIVE LABORATORY
Demonstration Microbiology Laboratory · DEMO-LAB
SYNC OFFLINE
Settings
GOVERNANCE
Audit trail
Inspect traceable local operations across configuration, imports, records, exports, federation and optional AI services.
Refresh
Help
74 LOADED EVENTS
73 SUCCESSFUL
1 WARNINGS / ERRORS
Search operation, summary or details…
All outcomes
74 events shown
TIME
OPERATION
SUMMARY
OUTCOME
DETAILS
14 Aug 2026, 11:56 am
ipc.error
llm:models: AI network access is off
ERROR
Inspect
14 Aug 2026, 11:56 am
breakpoints.eucast-update
950 EUCAST row(s) staged from network (15.0)
OK
14 Aug 2026, 11:56 am
breakpoint.stage
EUCAST Breakpoint Tables v15.0: 950 row(s) staged inactive
14 Aug 2026, 11:55 am
one-health.capture
environment: 9f9fc90c-6fc5-4467-aa57-16af0ddf0a08
veterinary: 65b048c9-cdf2-468f-9e65-ee3b9213a1ca
amc: f143cca1-f573-4c01-9fb0-636afdbb8e3b
human_quality: d12f25b0-8679-4da4-b7fa-5a95f558a84b
one-health.user.bootstrap
demo.admin: first administrator
isolate.save
Isolate 60

Part II — Central web portal

13. Deploy the server

The portal is Django with Postgres and Redis, shipped as a Docker Compose stack.

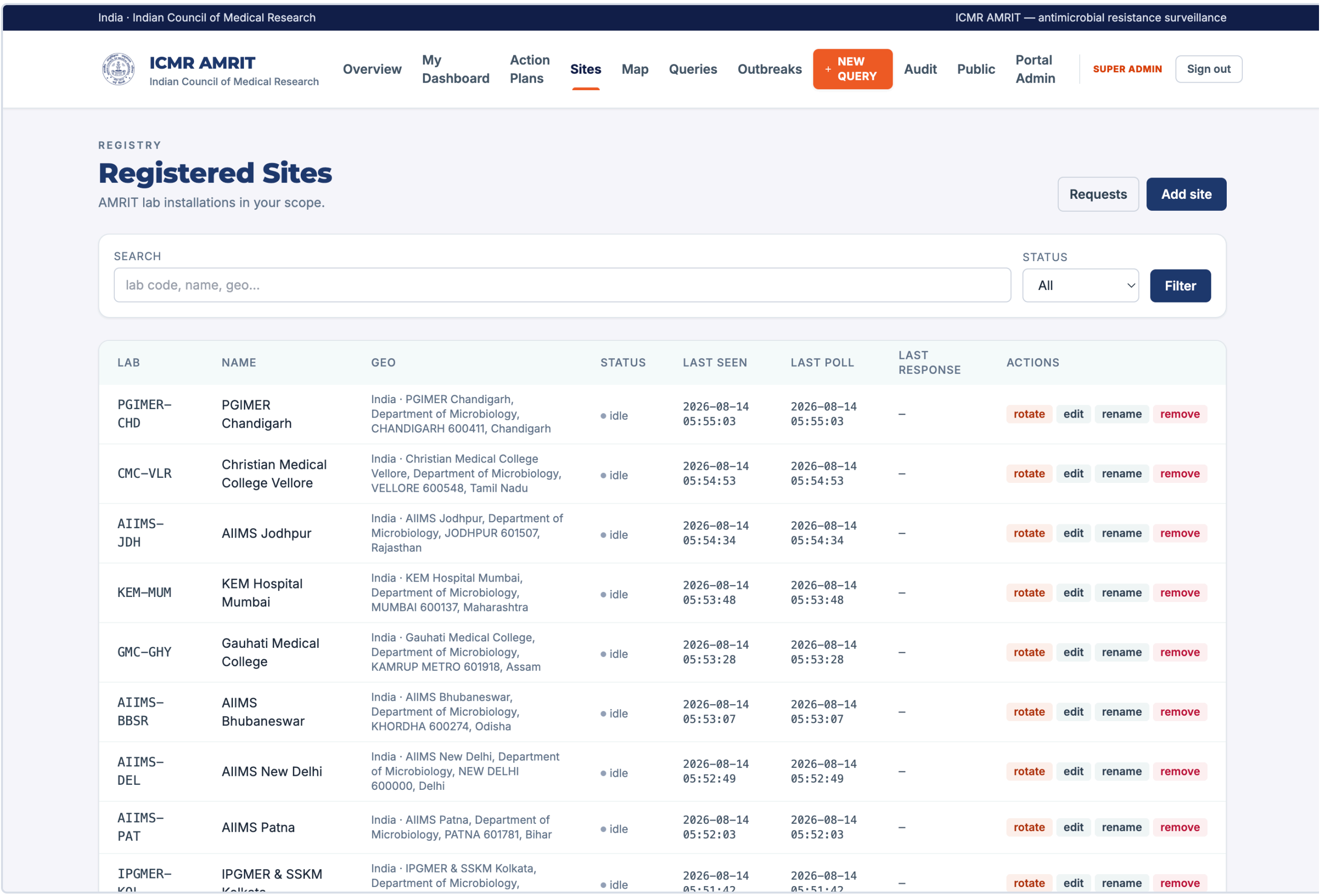
```
cd server/amrit_central_server
cp .env.example .env
# set at least: DJANGO_SECRET_KEY, DJANGO_ALLOWED_HOSTS, POSTGRES_PASSWORD,
#               AMRIT_COUNTRY_PROFILE (e.g. IN), AMRIT_ENROLMENT_SECRET
docker compose up -d --build
docker compose run --rm web python manage.py bootstrap   # first administrator
```

The entrypoint migrates, collects static assets and — only when `AMRIT_SEED_DEMO=1` — seeds the demonstration pack for the configured country. A country with no pack is reported and skipped, never filled with another country's hospitals.

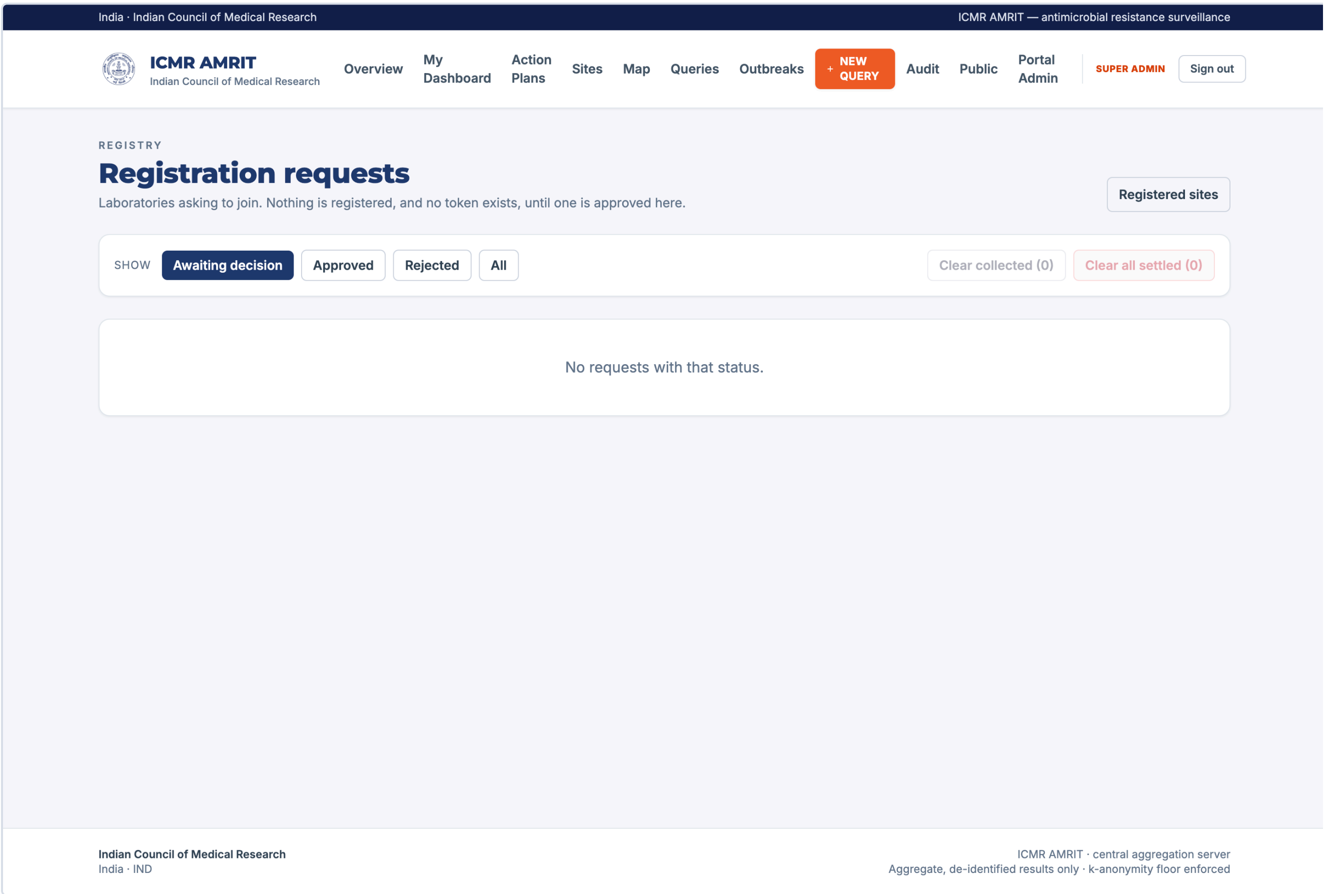
The stack publishes on loopback only. Nothing in it terminates TLS; put a reverse proxy in front before exposing it, because both operator sign-in and site enrolment carry credentials.

14. The site registry and enrolment

Registry → Sites lists every laboratory, its administrative unit, when it was last seen and whether it is online.

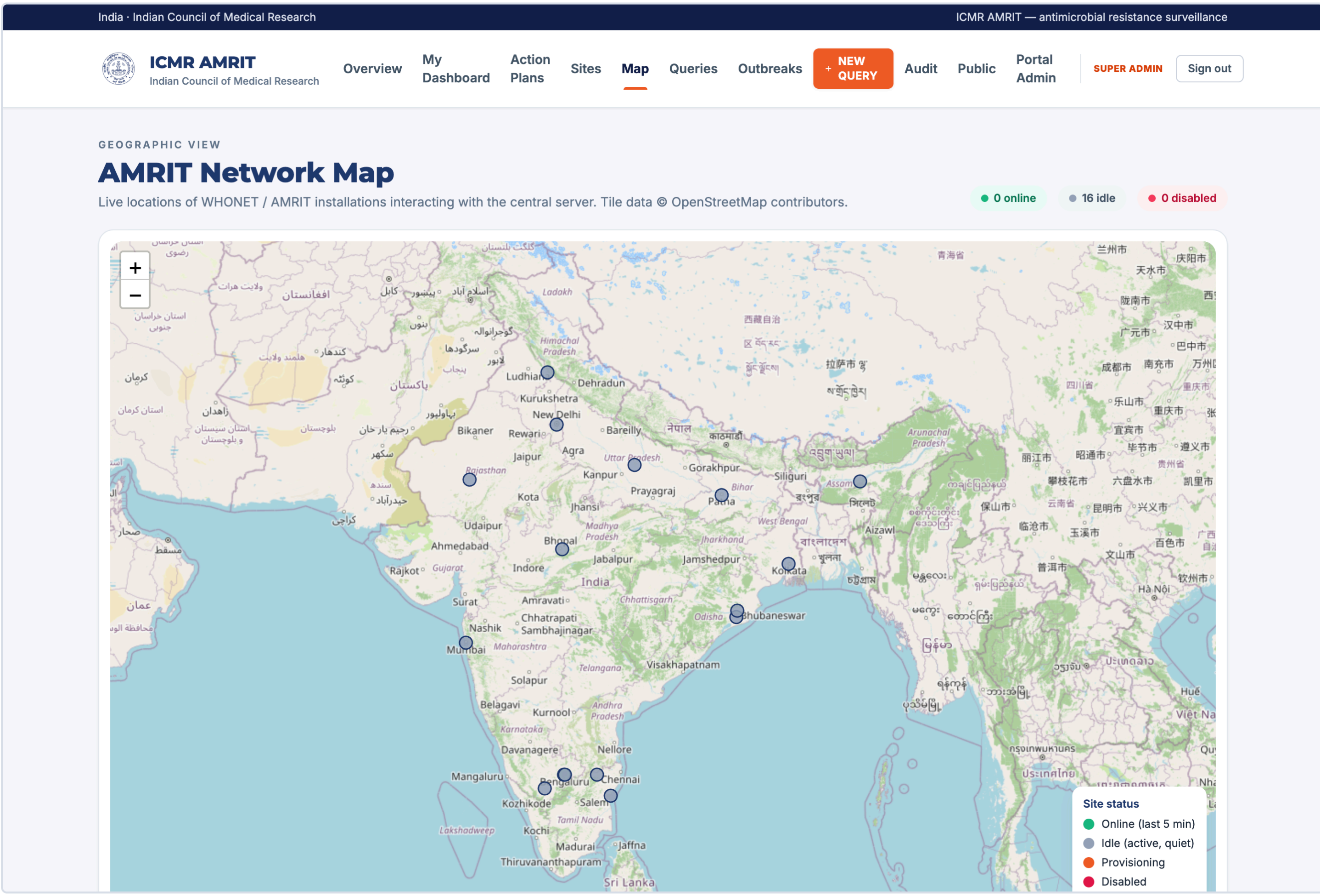


Registry → Requests is the enrolment queue: laboratories that have asked to join and are waiting for a decision. Approving creates the site and shows its site token exactly once.



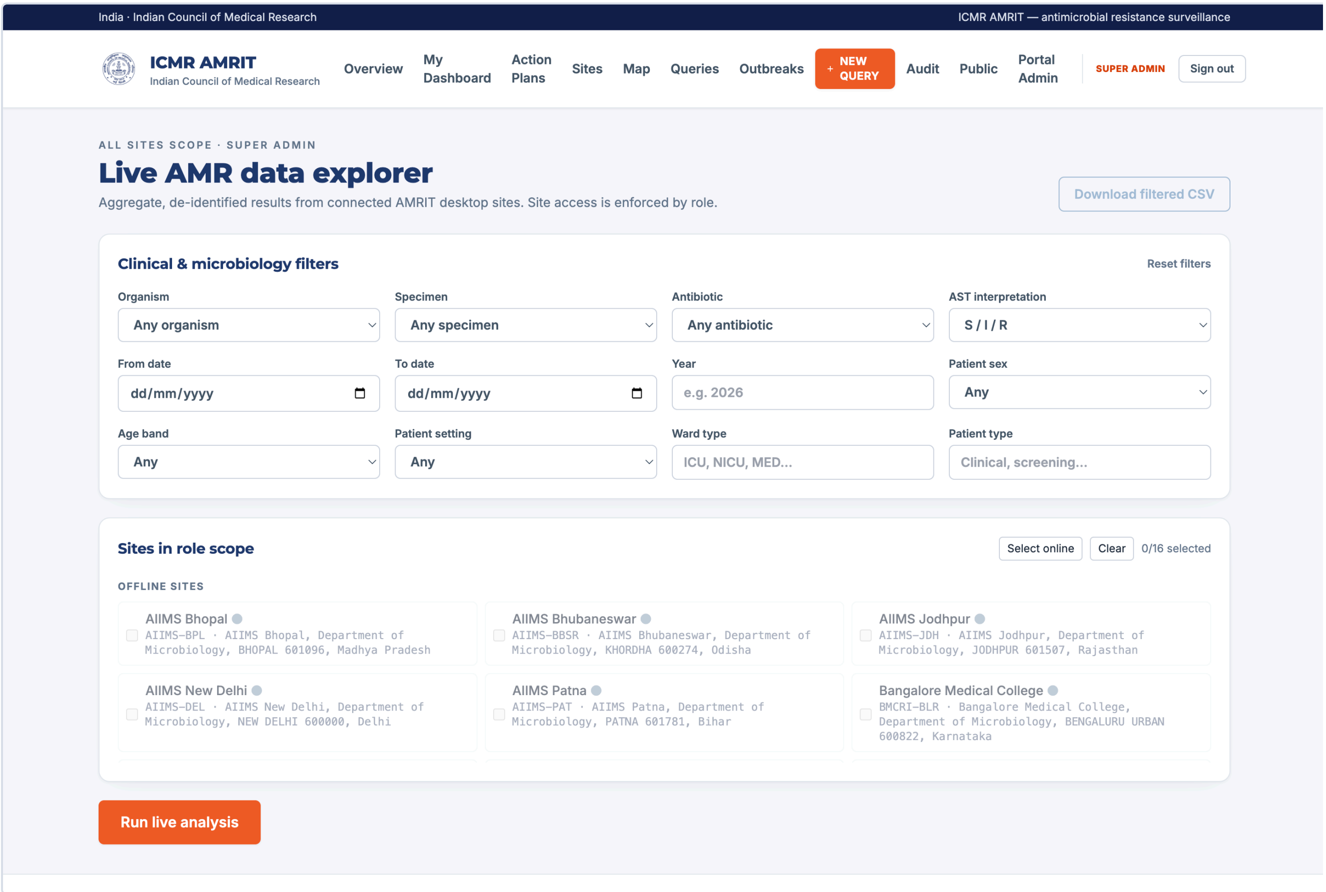
Both credentials can be reset independently, and each reset stops that site syncing until the new value reaches it — which the screen says before the button is pressed.

Registry → Map places every consenting site.



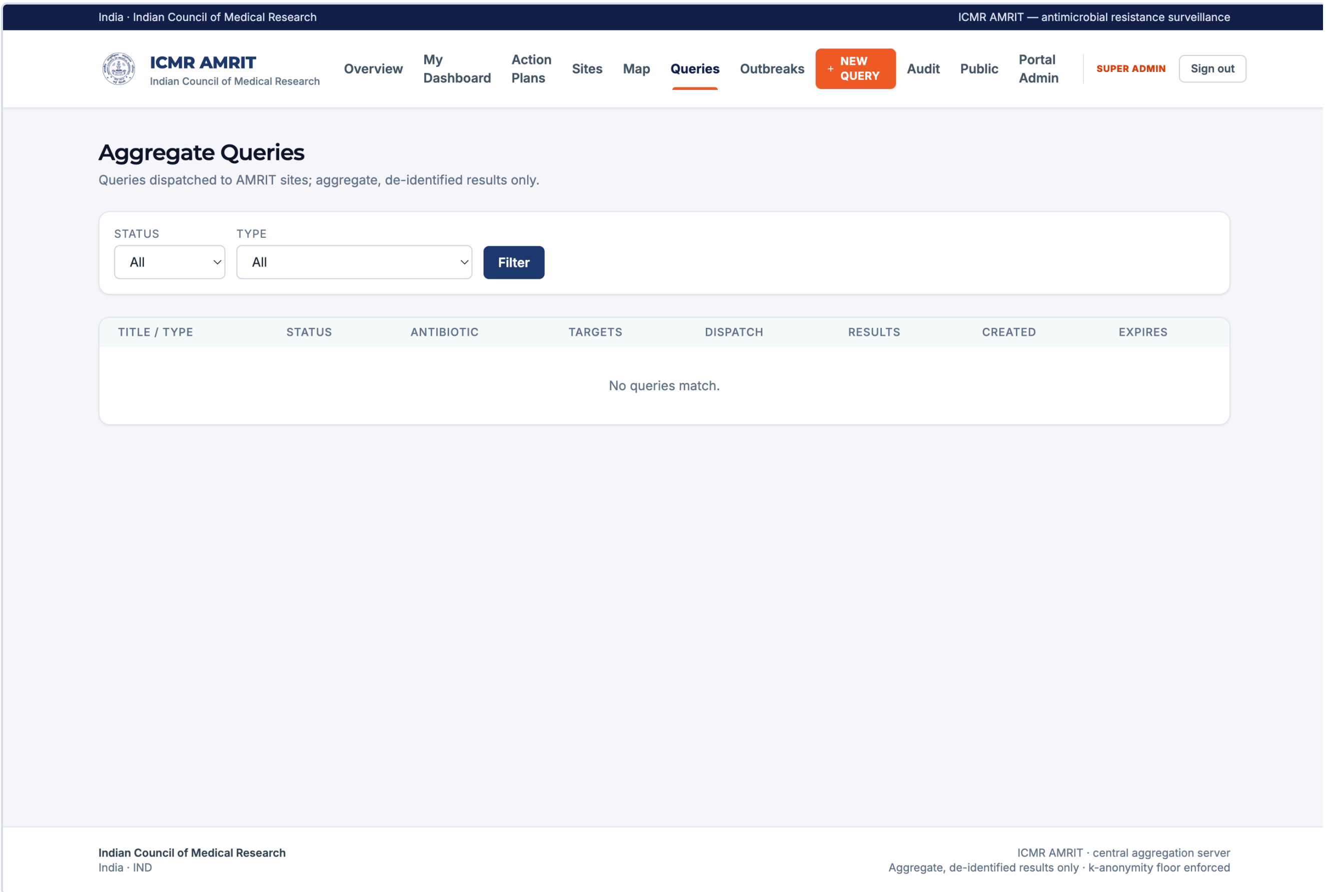
15. Asking questions

Queries → New query composes a question and targets it: all sites, an administrative unit, or named laboratories.



Query type	Answer
`isolate_count`	How many isolates match the filters
`organism_distribution`	Counts by organism
`specimen_distribution`	Counts by specimen type
`resistance_rate`	Numerator, denominator and percentage for an organism/agent pair
`measure_bundle`	The same as a FHIR MeasureReport
`cluster_scan`	Daily case counts by site and phenotype, for outbreak detection
`heartbeat`	Presence only

Queries expire (`AMRIT_QUERY_TTL_SECONDS`), and each site's answer is recorded separately, so a partial answer is visibly partial.



16. Dashboards, by stakeholder

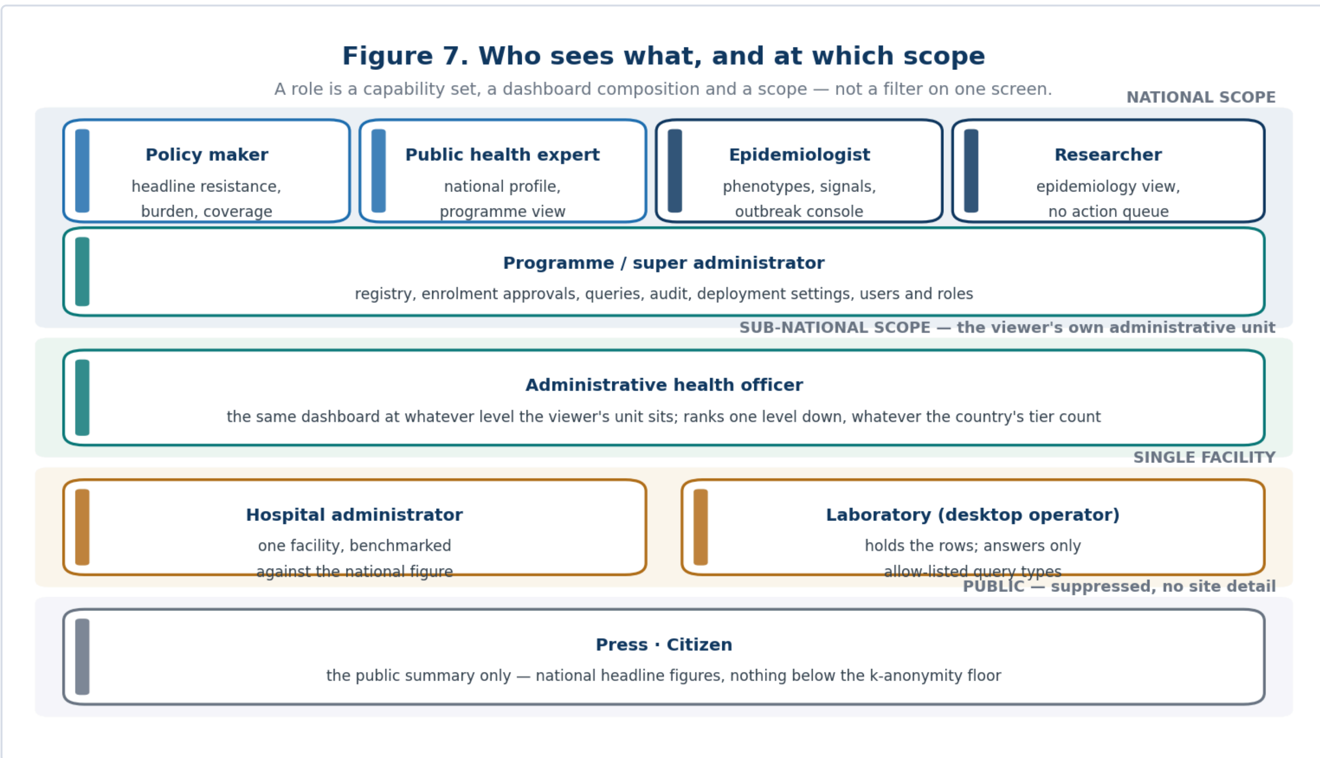
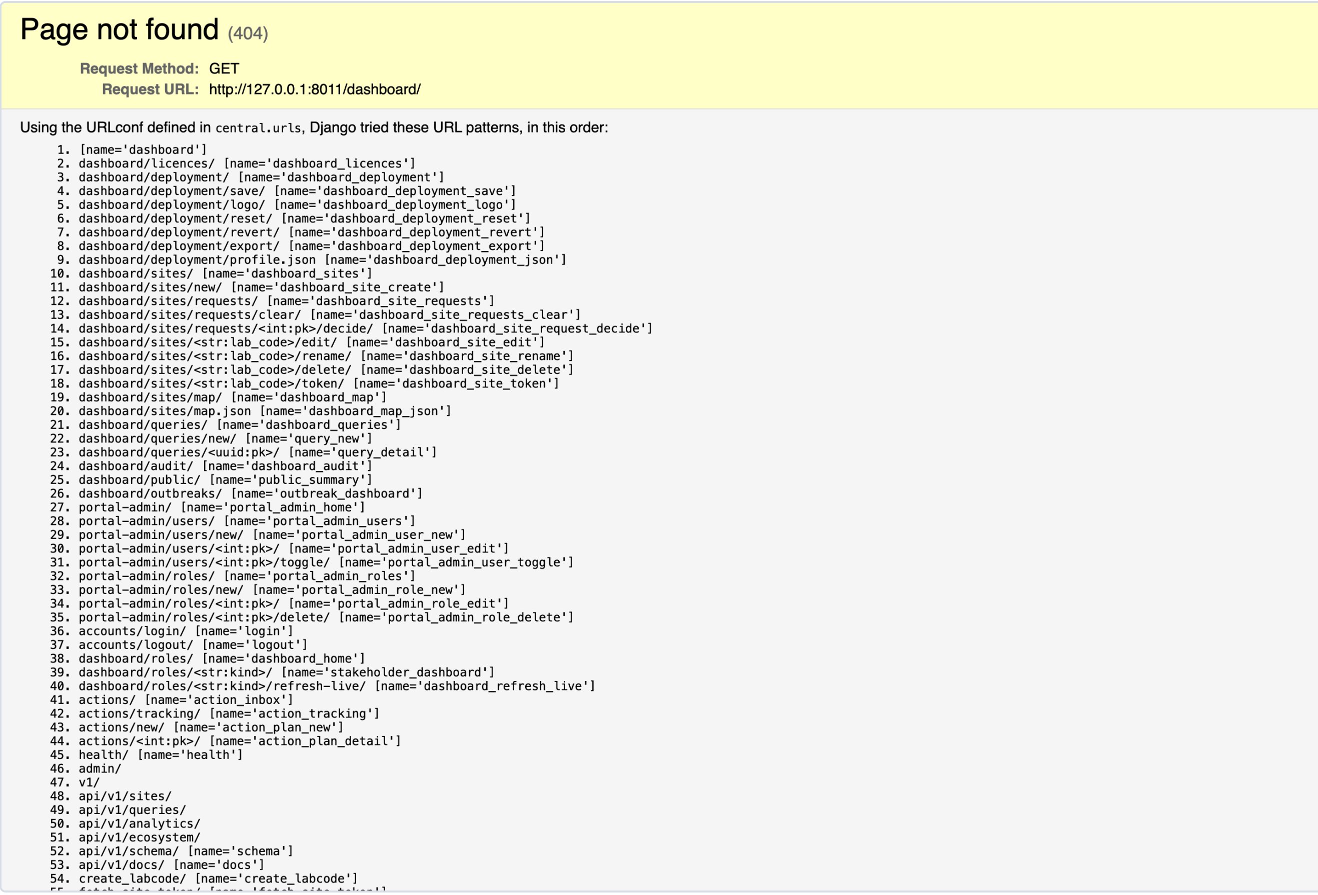


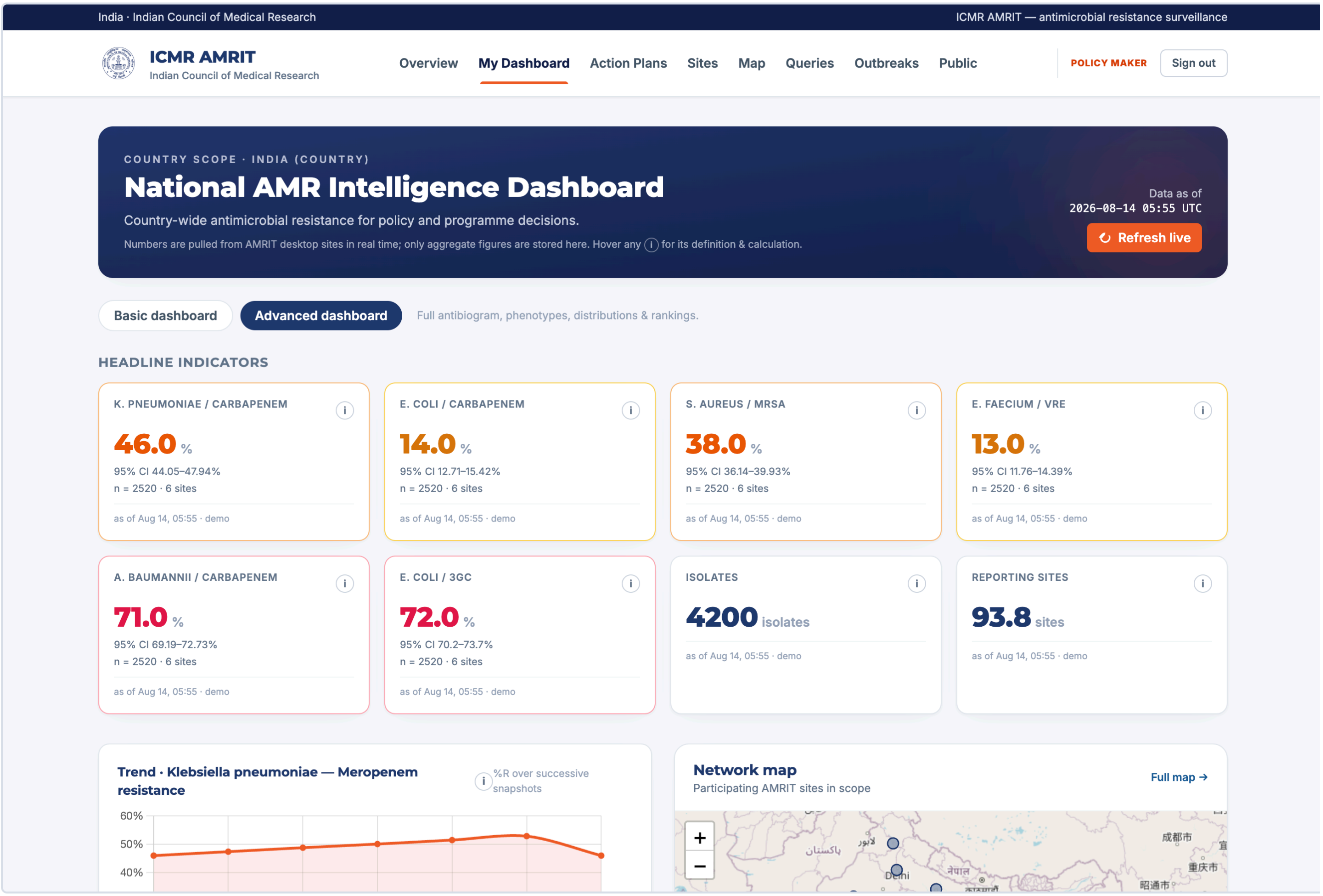
Figure 7 — who sees what, and at which scope.

A role is not a filter on one screen. Each stakeholder gets a different composition at a different scope, because the question each is asking is different.

Policy maker — national scope. Headline resistance, burden, coverage, trend.



The advanced section opens the full resistance matrix and phenotype panels.



Epidemiologist — national scope with signals. Priority-pathogen phenotypes (CRE, MDR), anomaly signals and the outbreak console.

Page not found (404)

Request Method: GET
Request URL: http://127.0.0.1:8011/dashboard/

Using the URLconf defined in `central.urls`, Django tried these URL patterns, in this order:

```
1. [name='dashboard']
2. dashboard/licences/ [name='dashboard_licences']
3. dashboard/deployment/ [name='dashboard_deployment']
4. dashboard/deployment/save/ [name='dashboard_deployment_save']
5. dashboard/deployment/logo/ [name='dashboard_deployment_logo']
6. dashboard/deployment/reset/ [name='dashboard_deployment_reset']
7. dashboard/deployment/revert/ [name='dashboard_deployment_revert']
8. dashboard/deployment/export/ [name='dashboard_deployment_export']
9. dashboard/deployment/profile.json [name='dashboard_deployment_json']
10. dashboard/sites/ [name='dashboard_sites']
11. dashboard/sites/new/ [name='dashboard_site_create']
12. dashboard/sites/requests/ [name='dashboard_site_requests']
13. dashboard/sites/requests/clear/ [name='dashboard_site_requests_clear']
14. dashboard/sites/requests/<int:pk>/decide/ [name='dashboard_site_request_decide']
15. dashboard/sites/<str:lab_code>/edit/ [name='dashboard_site_edit']
16. dashboard/sites/<str:lab_code>/rename/ [name='dashboard_site_rename']
17. dashboard/sites/<str:lab_code>/delete/ [name='dashboard_site_delete']
18. dashboard/sites/<str:lab_code>/token/ [name='dashboard_site_token']
19. dashboard/sites/map/ [name='dashboard_map']
20. dashboard/sites/map.json [name='dashboard_map_json']
21. dashboard/queries/ [name='dashboard_queries']
22. dashboard/queries/new/ [name='query_new']
23. dashboard/queries/<uuid:pk>/ [name='query_detail']
24. dashboard/audit/ [name='dashboard_audit']
25. dashboard/public/ [name='public_summary']
26. dashboard/outbreaks/ [name='outbreak_dashboard']
27. portal-admin/ [name='portal_admin_home']
28. portal-admin/users/ [name='portal_admin_users']
29. portal-admin/users/new/ [name='portal_admin_user_new']
30. portal-admin/users/<int:pk>/ [name='portal_admin_user_edit']
31. portal-admin/users/<int:pk>/toggle/ [name='portal_admin_user_toggle']
32. portal-admin/roles/ [name='portal_admin_roles']
33. portal-admin/roles/new/ [name='portal_admin_role_new']
34. portal-admin/roles/<int:pk>/ [name='portal_admin_role_edit']
35. portal-admin/roles/<int:pk>/delete/ [name='portal_admin_role_delete']
36. accounts/login/ [name='login']
37. accounts/logout/ [name='logout']
38. dashboard/roles/ [name='dashboard_home']
39. dashboard/roles/<str:kind>/ [name='stakeholder_dashboard']
40. dashboard/roles/<str:kind>/refresh-live/ [name='dashboard_refresh_live']
41. actions/ [name='action_inbox']
42. actions/tracking/ [name='action_tracking']
43. actions/new/ [name='action_plan_new']
44. actions/<int:pk>/ [name='action_plan_detail']
45. health/ [name='health']
46. admin/
47. v1/
48. api/v1/sites/
49. api/v1/queries/
50. api/v1/analytics/
51. api/v1/ecosystem/
52. api/v1/schema/ [name='schema']
53. api/v1/docs/ [name='docs']
54. create_labcode/ [name='create_labcode']
```

Researcher. The epidemiology composition without the operational action queue.

Page not found (404)

Request Method: GET
Request URL: http://127.0.0.1:8011/dashboard/

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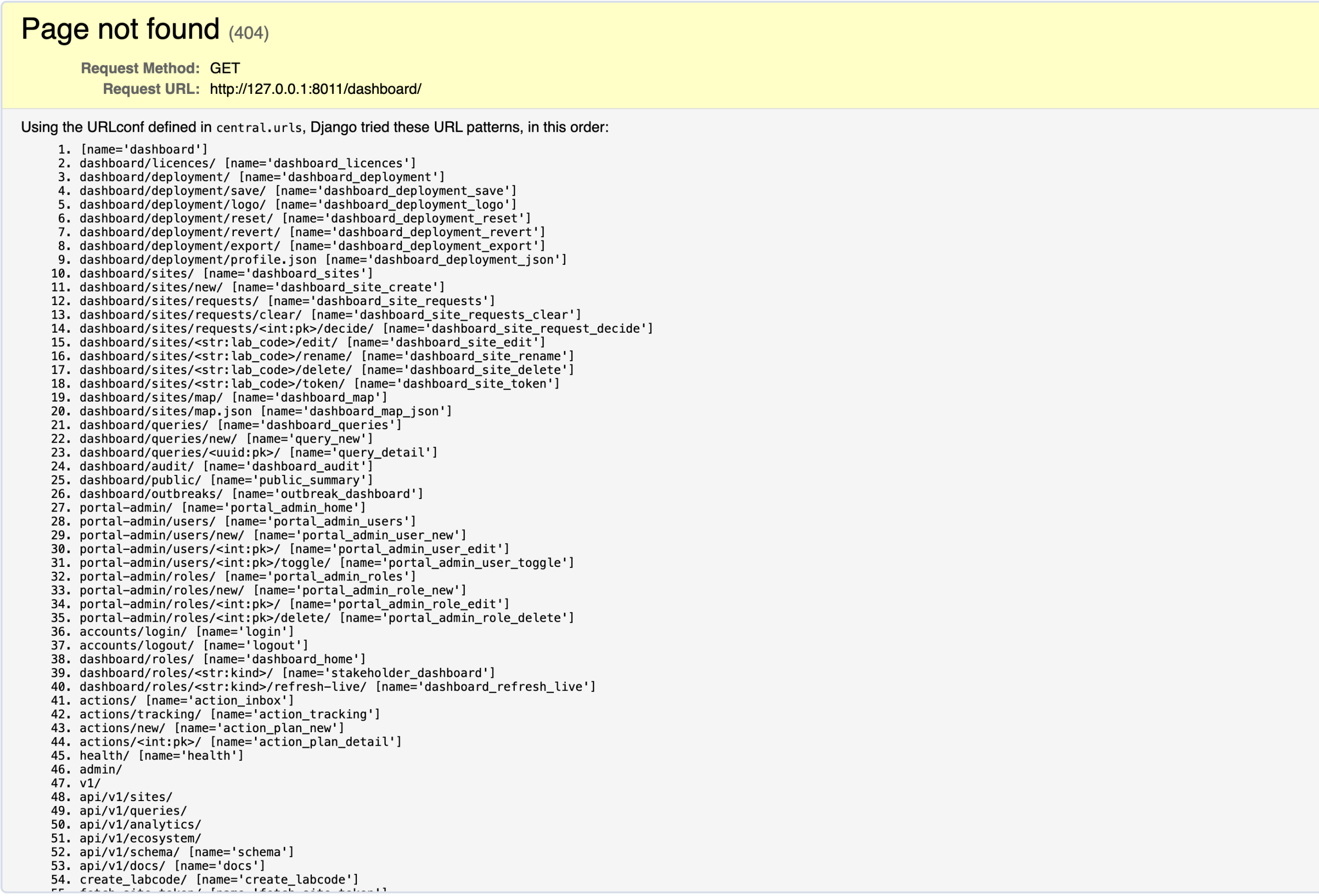
```
1. [name='dashboard']
2. dashboard/licences/ [name='dashboard_licences']
3. dashboard/deployment/ [name='dashboard_deployment']
4. dashboard/deployment/save/ [name='dashboard_deployment_save']
5. dashboard/deployment/logo/ [name='dashboard_deployment_logo']
6. dashboard/deployment/reset/ [name='dashboard_deployment_reset']
7. dashboard/deployment/revert/ [name='dashboard_deployment_revert']
8. dashboard/deployment/export/ [name='dashboard_deployment_export']
9. dashboard/deployment/profile.json [name='dashboard_deployment_json']
10. dashboard/sites/ [name='dashboard_sites']
11. dashboard/sites/new/ [name='dashboard_site_create']
12. dashboard/sites/requests/ [name='dashboard_site_requests']
13. dashboard/sites/requests/clear/ [name='dashboard_site_requests_clear']
14. dashboard/sites/requests/<int:pk>/decide/ [name='dashboard_site_request_decide']
15. dashboard/sites/<str:lab_code>/edit/ [name='dashboard_site_edit']
16. dashboard/sites/<str:lab_code>/rename/ [name='dashboard_site_rename']
17. dashboard/sites/<str:lab_code>/delete/ [name='dashboard_site_delete']
18. dashboard/sites/<str:lab_code>/token/ [name='dashboard_site_token']
19. dashboard/sites/map/ [name='dashboard_map']
20. dashboard/sites/map.json [name='dashboard_map_json']
21. dashboard/queries/ [name='dashboard_queries']
22. dashboard/queries/new/ [name='query_new']
23. dashboard/queries/<uuid:pk>/ [name='query_detail']
24. dashboard/audit/ [name='dashboard_audit']
25. dashboard/public/ [name='public_summary']
26. dashboard/outbreaks/ [name='outbreak_dashboard']
27. portal-admin/ [name='portal_admin_home']
28. portal-admin/users/ [name='portal_admin_users']
29. portal-admin/users/new/ [name='portal_admin_user_new']
30. portal-admin/users/<int:pk>/ [name='portal_admin_user_edit']
31. portal-admin/users/<int:pk>/toggle/ [name='portal_admin_user_toggle']
32. portal-admin/roles/ [name='portal_admin_roles']
33. portal-admin/roles/new/ [name='portal_admin_role_new']
34. portal-admin/roles/<int:pk>/ [name='portal_admin_role_edit']
35. portal-admin/roles/<int:pk>/delete/ [name='portal_admin_role_delete']
36. accounts/login/ [name='login']
37. accounts/logout/ [name='logout']
38. dashboard/roles/ [name='dashboard_home']
39. dashboard/roles/<str:kind>/ [name='stakeholder_dashboard']
40. dashboard/roles/<str:kind>/refresh-live/ [name='dashboard_refresh_live']
41. actions/ [name='action_inbox']
42. actions/tracking/ [name='action_tracking']
43. actions/new/ [name='action_plan_new']
44. actions/<int:pk>/ [name='action_plan_detail']
45. health/ [name='health']
46. admin/
47. v1/
48. api/v1/sites/
49. api/v1/queries/
50. api/v1/analytics/
51. api/v1/ecosystem/
52. api/v1/schema/ [name='schema']
53. api/v1/docs/ [name='docs']
54. create_labcode/ [name='create_labcode']
```

Public health expert — national scope.



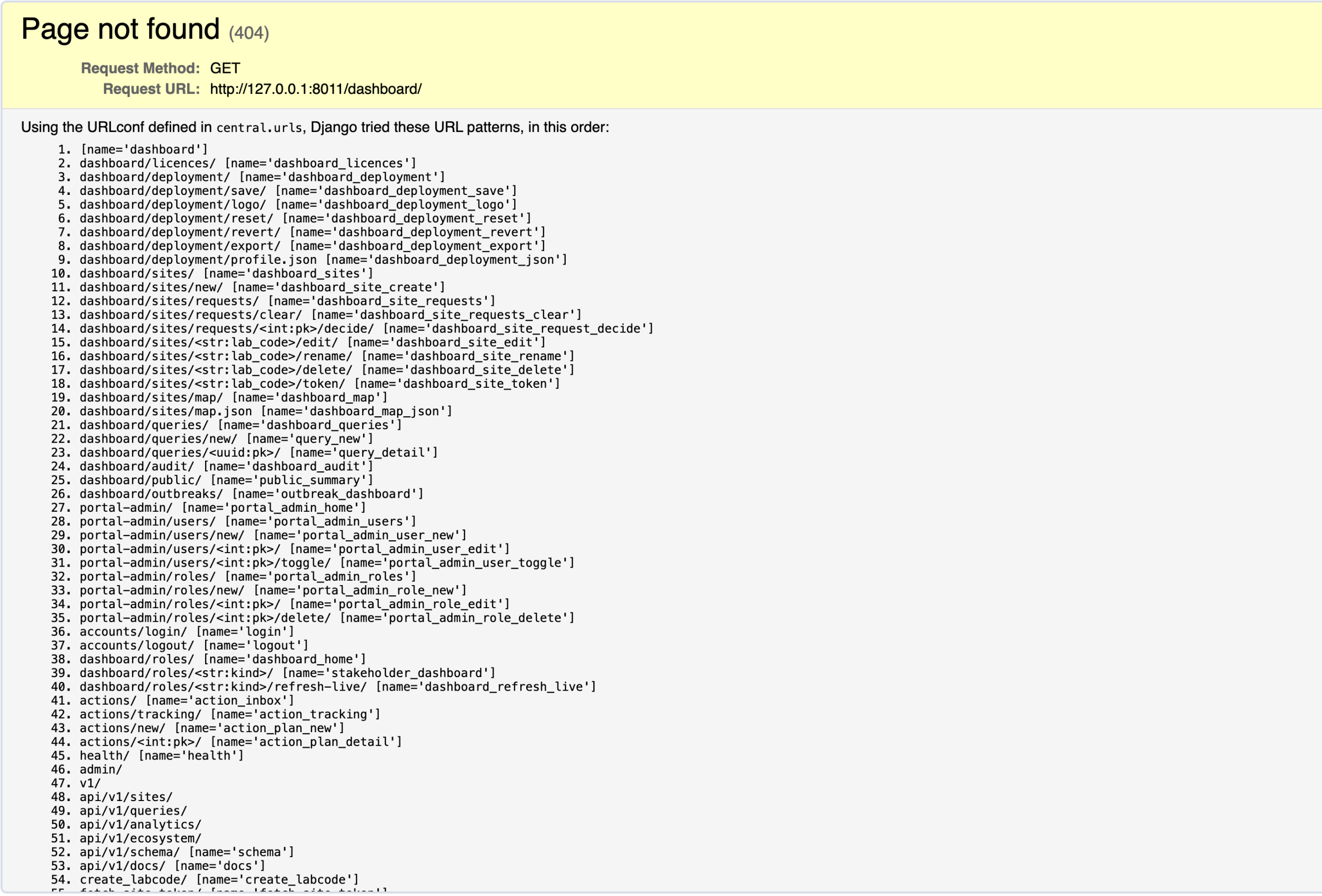
Page not found (404)

Request Method: GET
Request URL: http://127.0.0.1:8011/dashboard/

Using the URLconf defined in `central.urls`, Django tried these URL patterns, in this order:

```
1. [name='dashboard']
2. dashboard/licences/ [name='dashboard_licences']
3. dashboard/deployment/ [name='dashboard_deployment']
4. dashboard/deployment/save/ [name='dashboard_deployment_save']
5. dashboard/deployment/logo/ [name='dashboard_deployment_logo']
6. dashboard/deployment/reset/ [name='dashboard_deployment_reset']
7. dashboard/deployment/revert/ [name='dashboard_deployment_revert']
8. dashboard/deployment/export/ [name='dashboard_deployment_export']
9. dashboard/deployment/profile.json [name='dashboard_deployment_json']
10. dashboard/sites/ [name='dashboard_sites']
11. dashboard/sites/new/ [name='dashboard_site_create']
12. dashboard/sites/requests/ [name='dashboard_site_requests']
13. dashboard/sites/requests/clear/ [name='dashboard_site_requests_clear']
14. dashboard/sites/requests/<int:pk>/decide/ [name='dashboard_site_request_decide']
15. dashboard/sites/<str:lab_code>/edit/ [name='dashboard_site_edit']
16. dashboard/sites/<str:lab_code>/rename/ [name='dashboard_site_rename']
17. dashboard/sites/<str:lab_code>/delete/ [name='dashboard_site_delete']
18. dashboard/sites/<str:lab_code>/token/ [name='dashboard_site_token']
19. dashboard/sites/map/ [name='dashboard_map']
20. dashboard/sites/map.json [name='dashboard_map_json']
21. dashboard/queries/ [name='dashboard_queries']
22. dashboard/queries/new/ [name='query_new']
23. dashboard/queries/<uuid:pk>/ [name='query_detail']
24. dashboard/audit/ [name='dashboard_audit']
25. dashboard/public/ [name='public_summary']
26. dashboard/outbreaks/ [name='outbreak_dashboard']
27. portal-admin/ [name='portal_admin_home']
28. portal-admin/users/ [name='portal_admin_users']
29. portal-admin/users/new/ [name='portal_admin_user_new']
30. portal-admin/users/<int:pk>/ [name='portal_admin_user_edit']
31. portal-admin/users/<int:pk>/toggle/ [name='portal_admin_user_toggle']
32. portal-admin/roles/ [name='portal_admin_roles']
33. portal-admin/roles/new/ [name='portal_admin_role_new']
34. portal-admin/roles/<int:pk>/ [name='portal_admin_role_edit']
35. portal-admin/roles/<int:pk>/delete/ [name='portal_admin_role_delete']
36. accounts/login/ [name='login']
37. accounts/logout/ [name='logout']
38. dashboard/roles/ [name='dashboard_home']
39. dashboard/roles/<str:kind>/ [name='stakeholder_dashboard']
40. dashboard/roles/<str:kind>/refresh-live/ [name='dashboard_refresh_live']
41. actions/ [name='action_inbox']
42. actions/tracking/ [name='action_tracking']
43. actions/new/ [name='action_plan_new']
44. actions/<int:pk>/ [name='action_plan_detail']
45. health/ [name='health']
46. admin/
47. v1/
48. api/v1/sites/
49. api/v1/queries/
50. api/v1/analytics/
51. api/v1/ecosystem/
52. api/v1/schema/ [name='schema']
53. api/v1/docs/ [name='docs']
54. create_labcode/ [name='create_labcode']
```

Administrative health officer — sub-national scope. The same dashboard at whatever level the viewer's own administrative unit sits, ranking one level down. A three-tier country needs three *units*, not three dashboards.



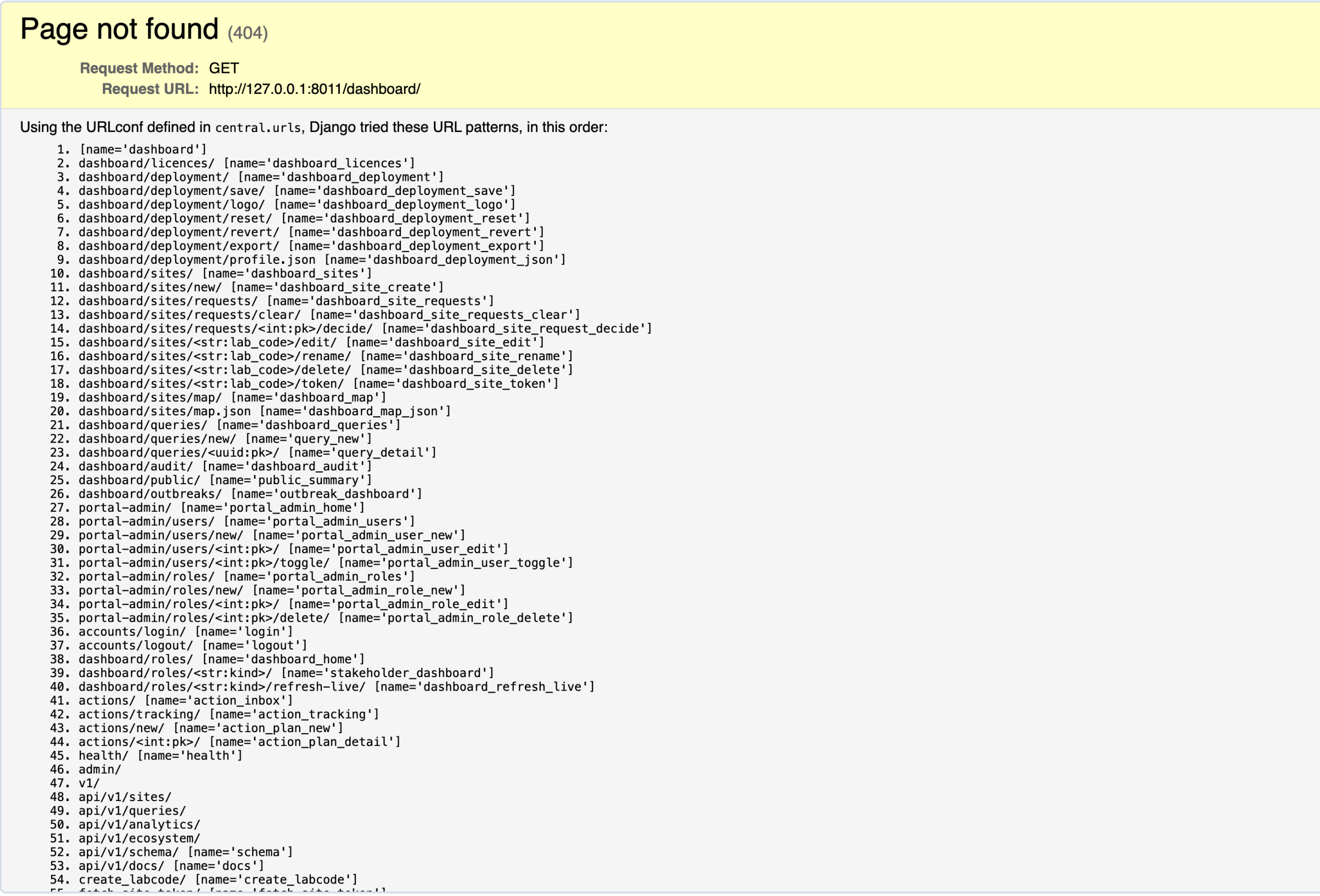
Page not found (404)

Request Method: GET
Request URL: http://127.0.0.1:8011/dashboard/

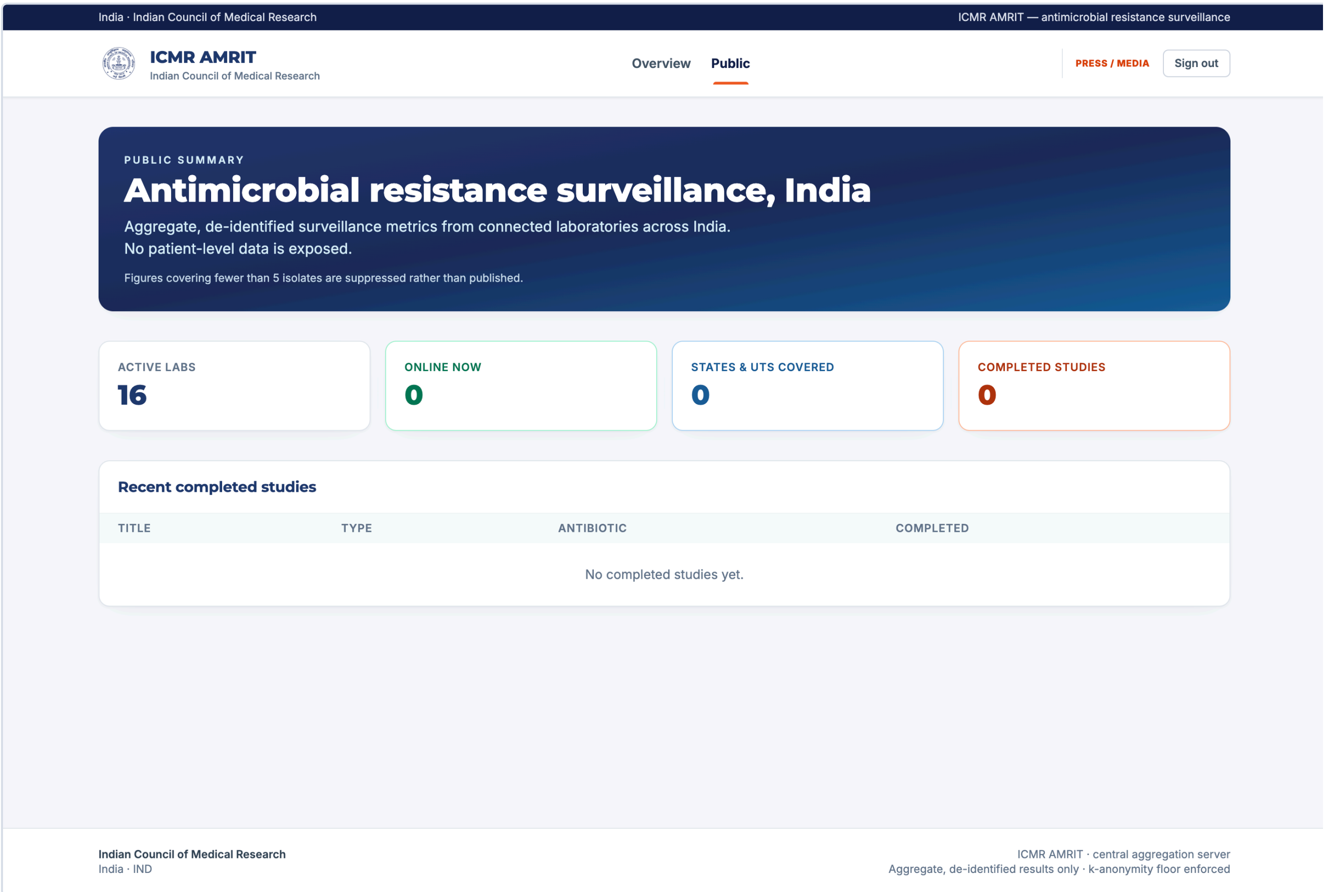
Using the URLconf defined in `central.urls`, Django tried these URL patterns, in this order:

```
 1. [name='dashboard']
 2. dashboard/licences/ [name='dashboard_licences']
 3. dashboard/deployment/ [name='dashboard_deployment']
 4. dashboard/deployment/save/ [name='dashboard_deployment_save']
 5. dashboard/deployment/logo/ [name='dashboard_deployment_logo']
 6. dashboard/deployment/reset/ [name='dashboard_deployment_reset']
 7. dashboard/deployment/revert/ [name='dashboard_deployment_revert']
 8. dashboard/deployment/export/ [name='dashboard_deployment_export']
 9. dashboard/deployment/profile.json [name='dashboard_deployment_json']
10. dashboard/sites/ [name='dashboard_sites']
11. dashboard/sites/new/ [name='dashboard_site_create']
12. dashboard/sites/requests/ [name='dashboard_site_requests']
13. dashboard/sites/requests/clear/ [name='dashboard_site_requests_clear']
14. dashboard/sites/requests/<int:pk>/decide/ [name='dashboard_site_request_decide']
15. dashboard/sites/<str:lab_code>/edit/ [name='dashboard_site_edit']
16. dashboard/sites/<str:lab_code>/rename/ [name='dashboard_site_rename']
17. dashboard/sites/<str:lab_code>/delete/ [name='dashboard_site_delete']
18. dashboard/sites/<str:lab_code>/token/ [name='dashboard_site_token']
19. dashboard/sites/map/ [name='dashboard_map']
20. dashboard/sites/map.json [name='dashboard_map_json']
21. dashboard/queries/ [name='dashboard_queries']
22. dashboard/queries/new/ [name='query_new']
23. dashboard/queries/<uuid:pk>/ [name='query_detail']
24. dashboard/audit/ [name='dashboard_audit']
25. dashboard/public/ [name='public_summary']
26. dashboard/outbreaks/ [name='outbreak_dashboard']
27. portal-admin/ [name='portal_admin_home']
28. portal-admin/users/ [name='portal_admin_users']
29. portal-admin/users/new/ [name='portal_admin_user_new']
30. portal-admin/users/<int:pk>/ [name='portal_admin_user_edit']
31. portal-admin/users/<int:pk>/toggle/ [name='portal_admin_user_toggle']
32. portal-admin/roles/ [name='portal_admin_roles']
33. portal-admin/roles/new/ [name='portal_admin_role_new']
34. portal-admin/roles/<int:pk>/ [name='portal_admin_role_edit']
35. portal-admin/roles/<int:pk>/delete/ [name='portal_admin_role_delete']
36. accounts/login/ [name='login']
37. accounts/logout/ [name='logout']
38. dashboard/roles/ [name='dashboard_home']
39. dashboard/roles/<str:kind>/ [name='stakeholder_dashboard']
40. dashboard/roles/<str:kind>/refresh-live/ [name='dashboard_refresh_live']
41. actions/ [name='action_inbox']
42. actions/tracking/ [name='action_tracking']
43. actions/new/ [name='action_plan_new']
44. actions/<int:pk>/ [name='action_plan_detail']
45. health/ [name='health']
46. admin/
47. v1/
48. api/v1/sites/
49. api/v1/queries/
50. api/v1/analytics/
51. api/v1/ecosystem/
52. api/v1/schema/ [name='schema']
53. api/v1/docs/ [name='docs']
54. create_labcode/ [name='create_labcode']
```

Hospital administrator — single-facility scope. One facility's profile, benchmarked against the national figure.



Press and citizen. The public summary only (§19).



Refresh live dispatches a fresh batch of queries to the in-scope sites, waits for the answers, and recomputes the snapshots for that scope (Figure 3). Historical results are never folded into a new refresh's numbers.

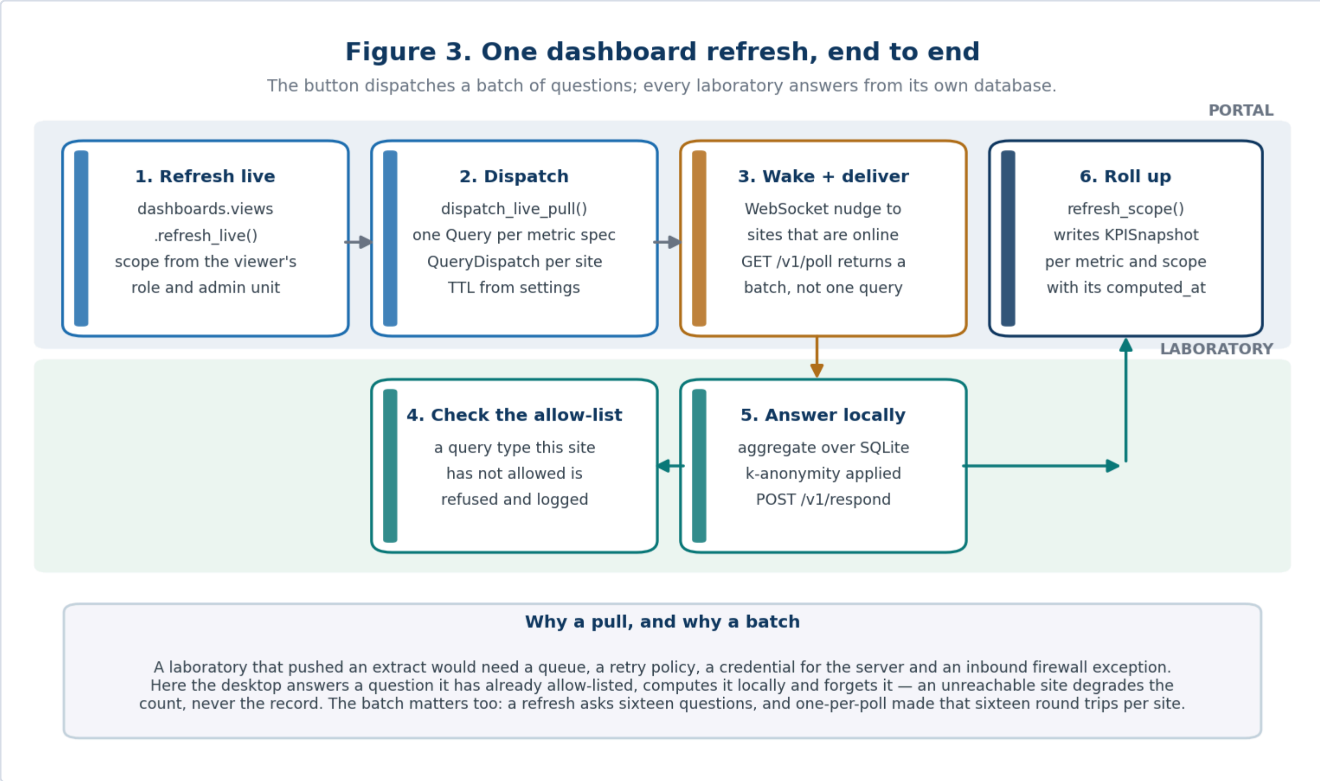
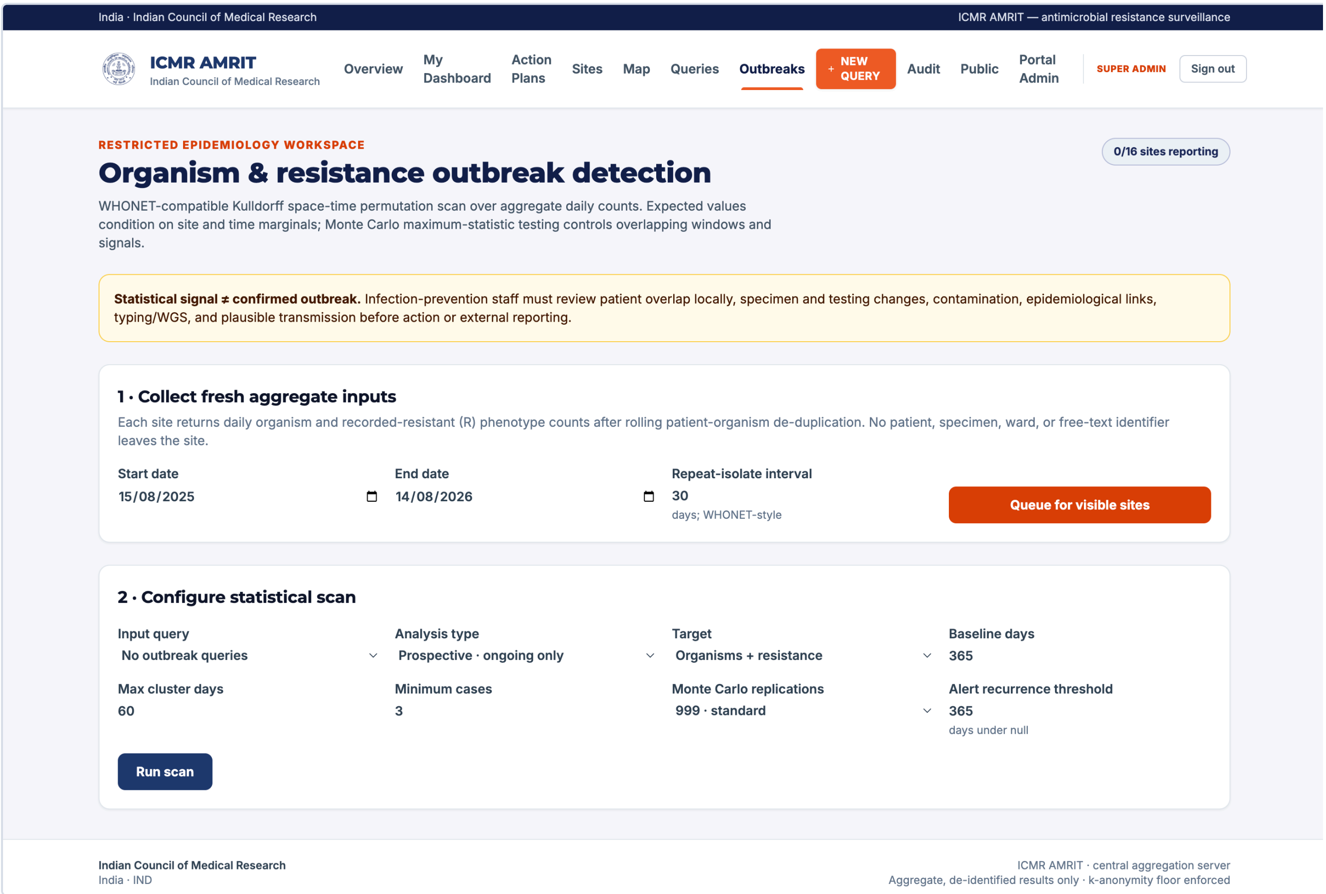


Figure 3 — one dashboard refresh, end to end.

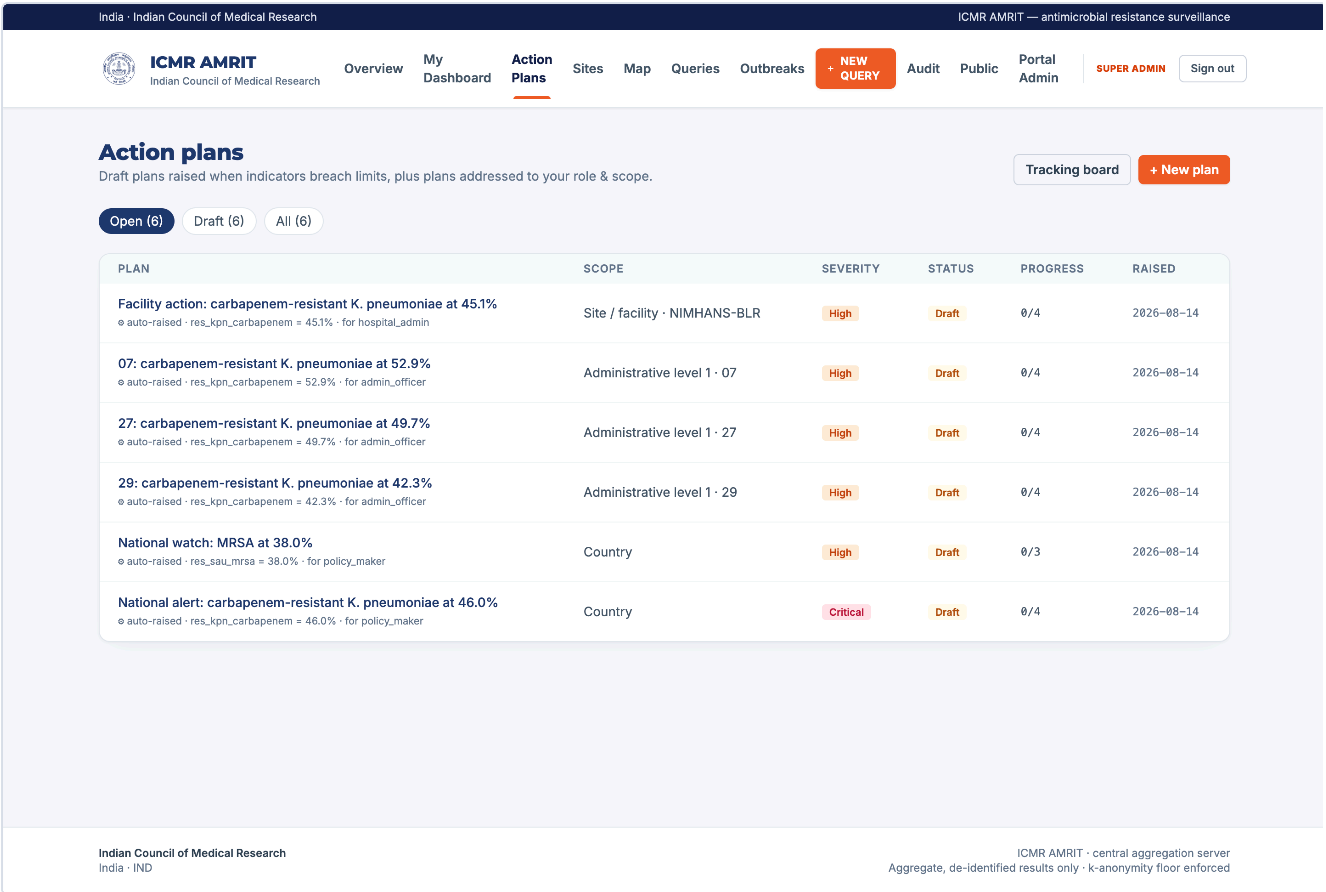
17. The outbreak console



The console runs the space–time permutation scan (§26) over `cluster_scan` aggregates and lists clusters with their window, sites, observed and expected counts, p-value and recurrence interval. Settings — baseline, maximum cluster length, minimum cases, permutations — are on the screen, because a signal without its parameters is not interpretable.

18. From a signal to an action plan

Threshold rules turn a metric crossing into a **draft** action plan — never an automatic instruction. A person edits, assigns and schedules it; tracking records what was done.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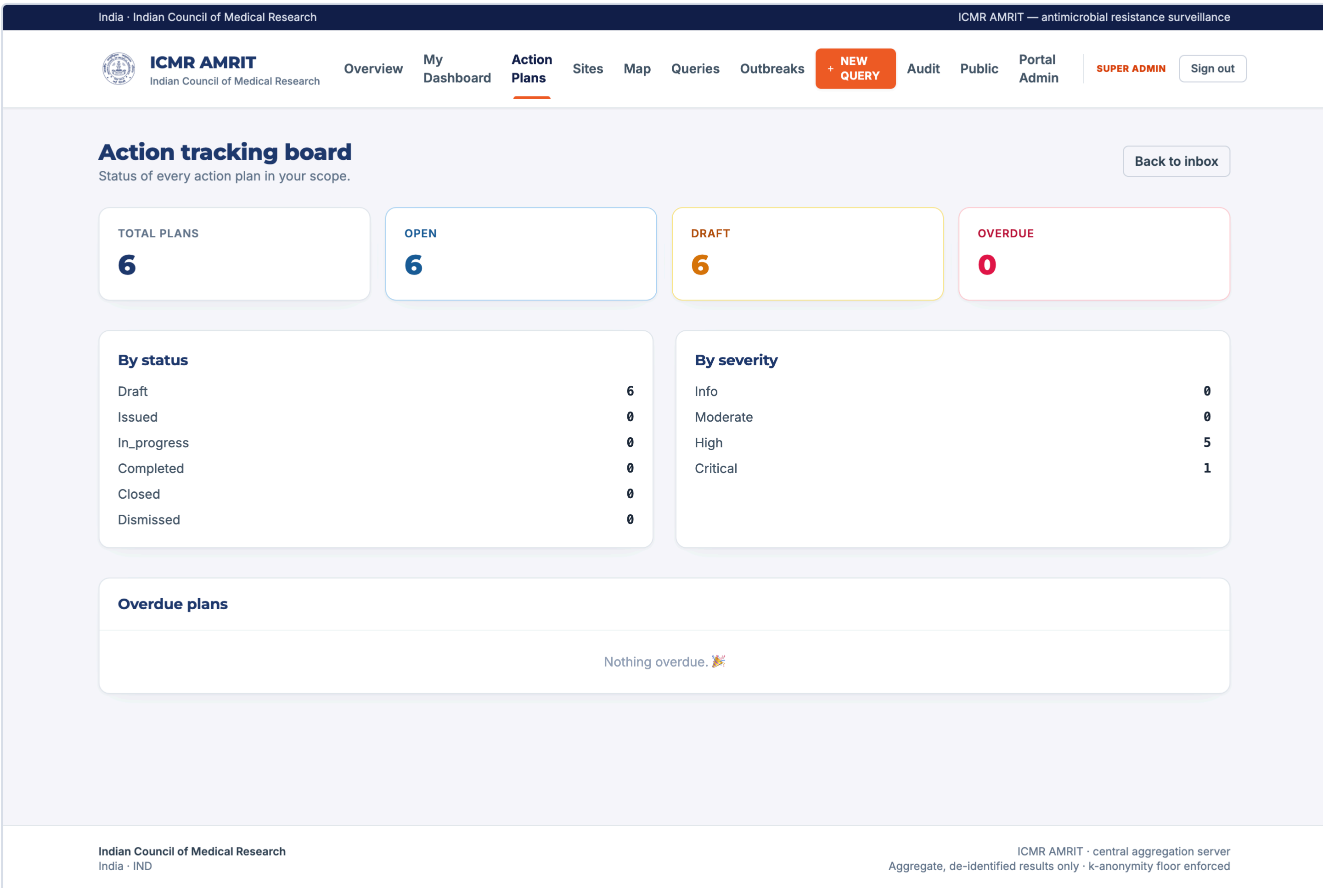
ICMR AMRIT — antimicrobial resistance surveillance
ICMR AMRIT
Indian Council of Medical Research
Overview
My Dashboard
Action Plans
Sites
Map
Queries
Outbreaks
+ NEW QUERY
Audit
Public
Portal Admin
SUPER ADMIN
Sign out
Action plans
Draft plans raised when indicators breach limits, plus plans addressed to your role & scope.
Tracking board
+ New plan
Open (6)
Draft (6)
All (6)
PLAN
SCOPE
SEVERITY
STATUS
PROGRESS
RAISED
Facility action: carbapenem-resistant K. pneumoniae at 45.1%
auto-raised · res_kpn_carbapenem = 45.1% · for hospital_admin
Site / facility · NIMHANS-BLR
High
Draft
0/4
2026-08-14
07: carbapenem-resistant K. pneumoniae at 52.9%
auto-raised · res_kpn_carbapenem = 52.9% · for admin_officer
Administrative level 1 · 07
High
Draft
0/4
2026-08-14
27: carbapenem-resistant K. pneumoniae at 49.7%
auto-raised · res_kpn_carbapenem = 49.7% · for admin_officer
Administrative level 1 · 27
High
Draft
0/4
2026-08-14
29: carbapenem-resistant K. pneumoniae at 42.3%
auto-raised · res_kpn_carbapenem = 42.3% · for admin_officer
Administrative level 1 · 29
High
Draft
0/4
2026-08-14
National watch: MRSA at 38.0%
auto-raised · res_sau_mrsa = 38.0% · for policy_maker
Country
High
Draft
0/3
2026-08-14
National alert: carbapenem-resistant K. pneumoniae at 46.0%
auto-raised · res_kpn_carbapenem = 46.0% · for policy_maker
Country
Critical
Draft
0/4
2026-08-14
Indian Council of Medical Research
India · IND
ICMR AMRIT · central aggregation server
Aggregate, de-identified results only · k-anonymity floor enforced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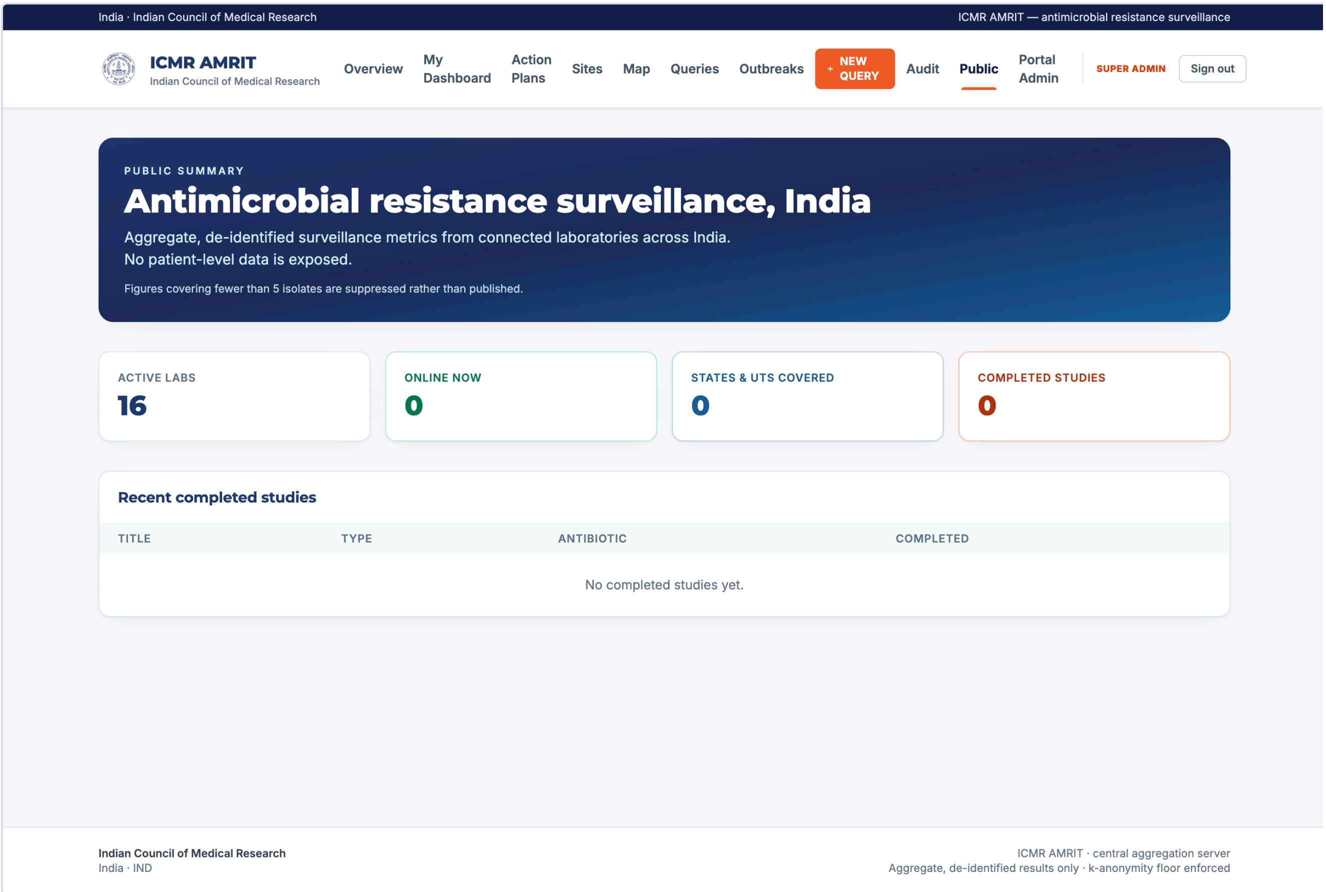
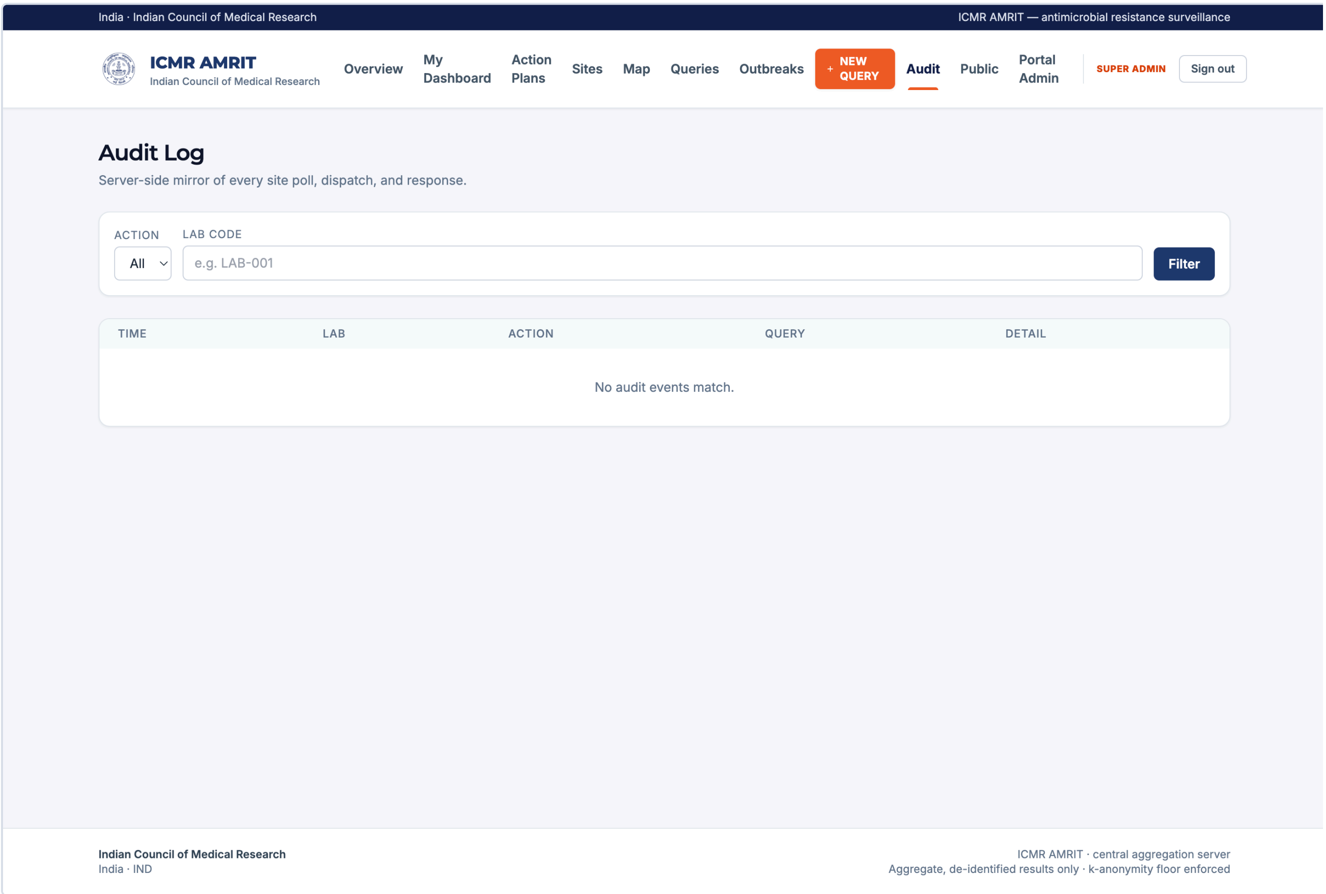
Overview
My Dashboard
Action Plans
Sites
Map
Queries
Outbreaks
+ NEW QUERY
Audit
Public
Portal Admin
SUPER ADMIN
Sign out
Action tracking board
Status of every action plan in your scope.
Back to inbox
TOTAL PLANS
6
OPEN
6
DRAFT
6
OVERDUE
0
By status
Draft 6
Issued 0
In_progress 0
Completed 0
Closed 0
Dismissed 0
By severity
Info 0
Moderate 0
High 5
Critical 1
Overdue plans
Nothing overdue. 🎉
Indian Council of Medical Research
India · IND
ICMR AMRIT · central aggregation server
Aggregate, de-identified results only · k-anonymity floor enforced

19. The public view

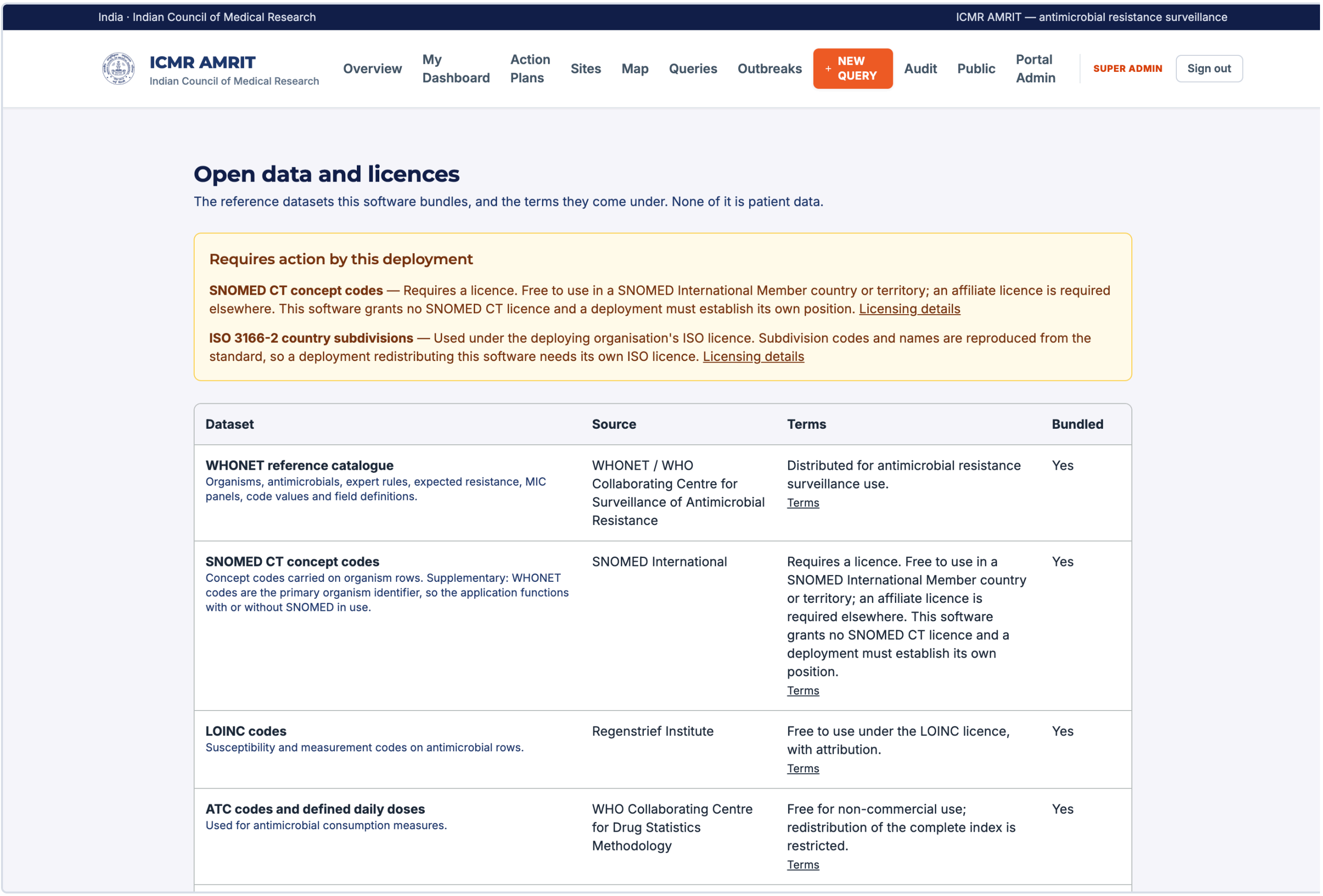
A deliberately reduced summary for press and citizens: national headline figures, no site-level detail, everything already suppressed below the k-anonymity floor.



20. Audit and governance



Every dispatch, answer, approval, token reset, rename and deletion is recorded with actor, time and detail. Licences lists every bundled dataset and its terms.



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ICMR AMRIT
Indian Council of Medical Research

Overview My Dashboard Action Plans Sites Map Queries Outbreaks + NEW QUERY Audit Public Portal Admin SUPER ADMIN Sign out

Open data and licences

The reference datasets this software bundles, and the terms they come under. None of it is patient data.

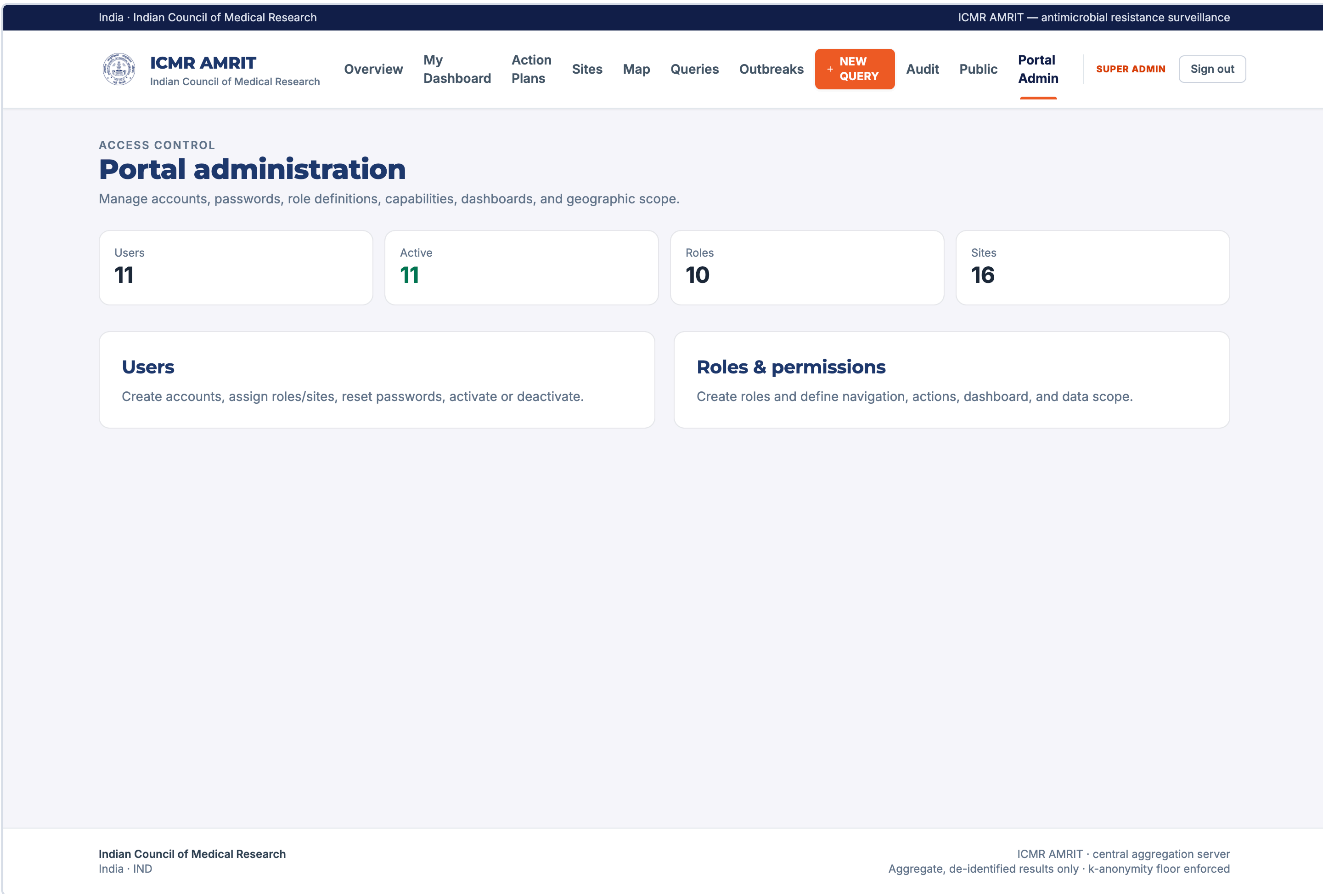
Requires action by this deployment

SNOMED CT concept codes — Requires a licence. Free to use in a SNOMED International Member country or territory; an affiliate licence is required elsewhere. This software grants no SNOMED CT licence and a deployment must establish its own position. Licensing details

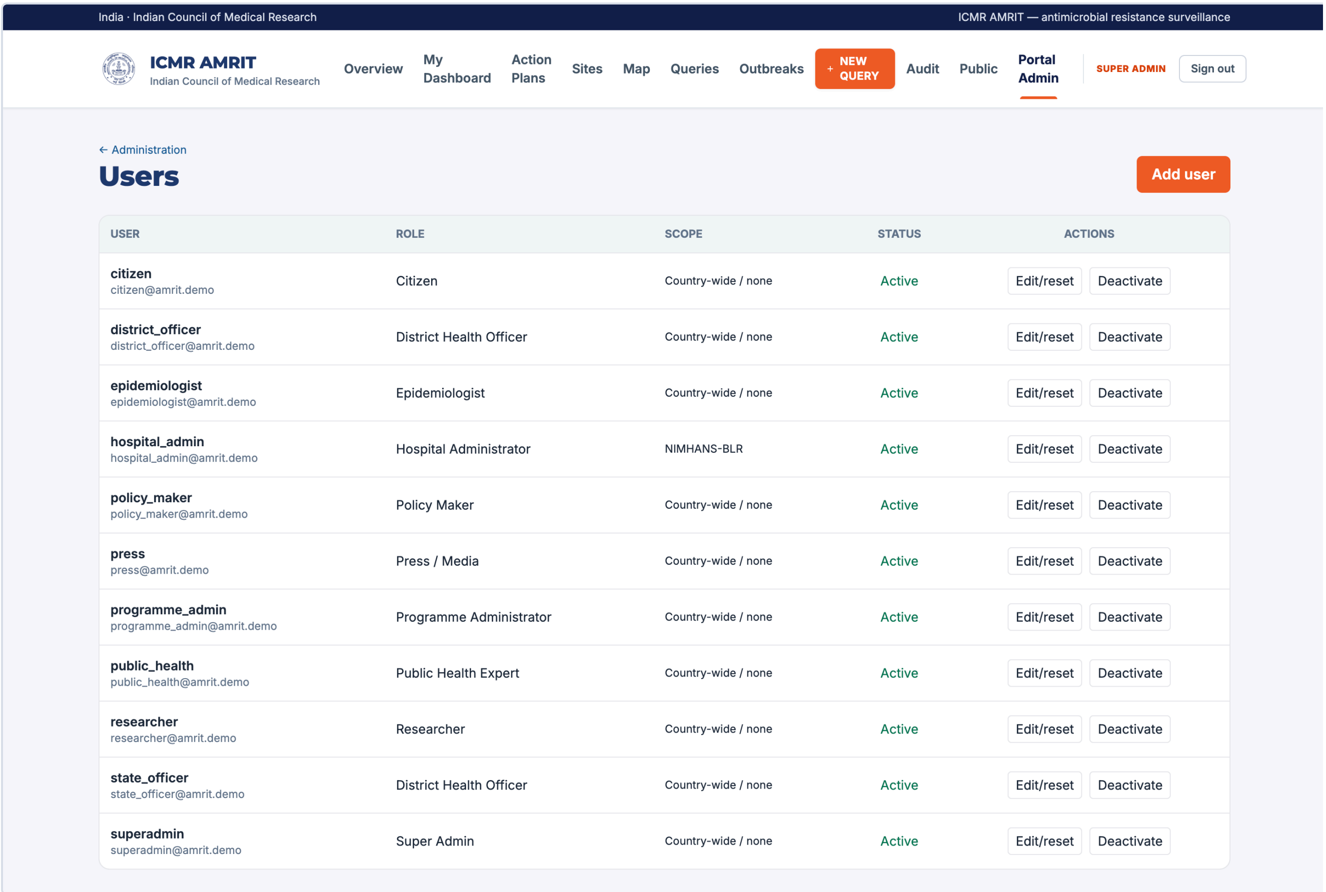
ISO 3166-2 country subdivisions — Used under the deploying organisation's ISO licence. Subdivision codes and names are reproduced from the standard, so a deployment redistributing this software needs its own ISO licence. Licensing details

Dataset	Source	Terms	Bundled
WHONET reference catalogue Organisms, antimicrobials, expert rules, expected resistance, MIC panels, code values and field definitions.	WHONET / WHO Collaborating Centre for Surveillance of Antimicrobial Resistance	Distributed for antimicrobial resistance surveillance use. Terms	Yes
SNOMED CT concept codes Concept codes carried on organism rows. Supplementary: WHONET codes are the primary organism identifier, so the application functions with or without SNOMED in use.	SNOMED International	Requires a licence. Free to use in a SNOMED International Member country or territory; an affiliate licence is required elsewhere. This software grants no SNOMED CT licence and a deployment must establish its own position. Terms	Yes
LOINC codes Susceptibility and measurement codes on antimicrobial rows.	Regenstrief Institute	Free to use under the LOINC licence, with attribution. Terms	Yes
ATC codes and defined daily doses Used for antimicrobial consumption measures.	WHO Collaborating Centre for Drug Statistics Methodology	Free for non-commercial use; redistribution of the complete index is restricted. Terms	Yes

21. Portal administration



Users and roles are managed in the portal itself. A role is a set of capabilities plus a dashboard kind plus a scope; a deployment can rename roles and change what each may see without a code change.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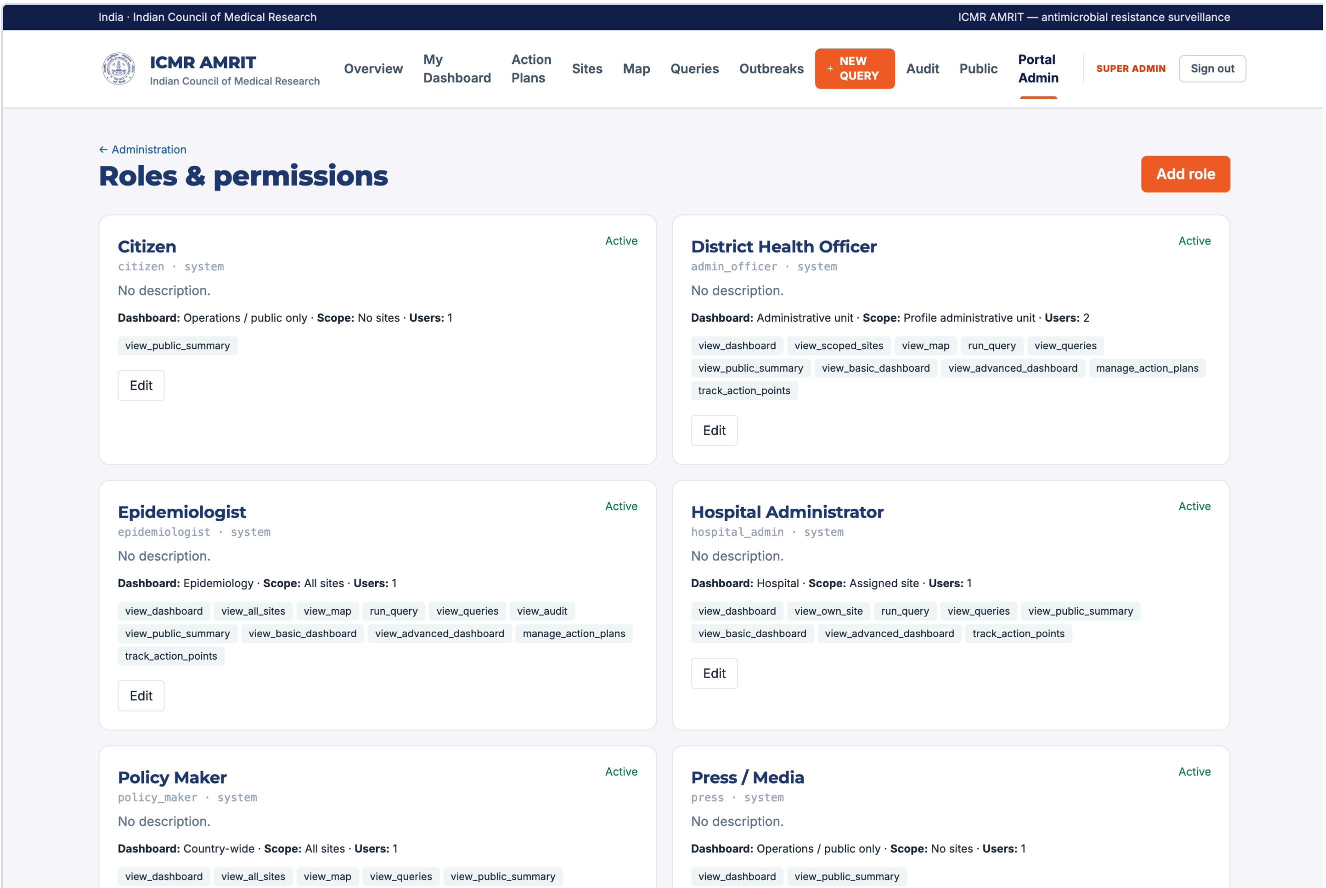
Outbreaks
+ NEW QUERY
Audit
Public
Portal Admin
SUPER ADMIN
Sign out
← Administration
Users
Add user
USER
ROLE
SCOPE
STATUS
ACTIONS
citizen
citizen@amrit.demo
Citizen
Country-wide / none
Active
Edit/reset
Deactivate
district_officer
district_officer@amrit.demo
District Health Officer
Country-wide / none
Active
Edit/reset
Deactivate
epidemiologist
epidemiologist@amrit.demo
Epidemiologist
Country-wide / none
Active
Edit/reset
Deactivate
hospital_admin
hospital_admin@amrit.demo
Hospital Administrator
NIMHANS-BLR
Active
Edit/reset
Deactivate
policy_maker
policy_maker@amrit.demo
Policy Maker
Country-wide / none
Active
Edit/reset
Deactivate
press
press@amrit.demo
Press / Media
Country-wide / none
Active
Edit/reset
Deactivate
programme_admin
programme_admin@amrit.demo
Programme Administrator
Country-wide / none
Active
Edit/reset
Deactivate
public_health
public_health@amrit.demo
Public Health Expert
Country-wide / none
Active
Edit/reset
Deactivate
researcher
researcher@amrit.demo
Researcher
Country-wide / none
Active
Edit/reset
Deactivate
state_officer
state_officer@amrit.demo
District Health Officer
Country-wide / none
Active
Edit/reset
Deactivate
superadmin
superadmin@amrit.demo
Super Admin
Country-wide / none
Active
Edit/reset
Deactivate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Overview
My Dashboard
Action Plans
Sites
Map
Queries
Outbreaks
+ NEW QUERY
Audit
Public
Portal Admin
SUPER ADMIN
Sign out
← Administration
Roles & permissions
Add role
Citizen
Active
citizen · system
No description.
Dashboard: Operations / public only · Scope: No sites · Users: 1
view_public_summary
Edit
District Health Officer
Active
admin_officer · system
No description.
Dashboard: Administrative unit · Scope: Profile administrative unit · Users: 2
view_dashboard
view_scoped_sites
view_map
run_query
view_queries
view_public_summary
view_basic_dashboard
view_advanced_dashboard
manage_action_plans
track_action_points
Edit
Epidemiologist
Active
epidemiologist · system
No description.
Dashboard: Epidemiology · Scope: All sites · Users: 1
view_dashboard
view_all_sites
view_map
run_query
view_queries
view_audit
view_public_summary
view_basic_dashboard
view_advanced_dashboard
manage_action_plans
track_action_points
Edit
Hospital Administrator
Active
hospital_admin · system
No description.
Dashboard: Hospital · Scope: Assigned site · Users: 1
view_dashboard
view_own_site
run_query
view_queries
view_public_summary
view_basic_dashboard
view_advanced_dashboard
track_action_points
Edit
Policy Maker
Active
policy_maker · system
No description.
Dashboard: Country-wide · Scope: All sites · Users: 1
view_dashboard
view_all_sites
view_map
view_queries
view_public_summary
Press / Media
Active
press · system
No description.
Dashboard: Operations / public only · Scope: No sites · Users: 1
view_dashboard
view_public_summary

Part III — Methods and rationale

This part states, for each analytical choice: what the method is, what the alternatives were, and why this one was taken. It is the material a paper would cite.

22. Data model and standards

Records are stored in the WHONET code space (organism, antimicrobial and specimen codes) so data is exchangeable with the GLASS ecosystem without a mapping step. Interoperable output is produced on demand: FHIR `Observation` for results, `MeasureReport` for aggregates, HL7 v2 for hospital systems. Organism concepts carry SNOMED CT codes and antimicrobials carry ATC and WHO AWaRe classification, so an indicator can be defined in terms a programme already uses.

Why not a bespoke schema? Because the value of an AMR record is comparability. A national system whose codes are its own becomes an island, and the mapping cost is paid later, by somebody with less context.

23. Breakpoint interpretation

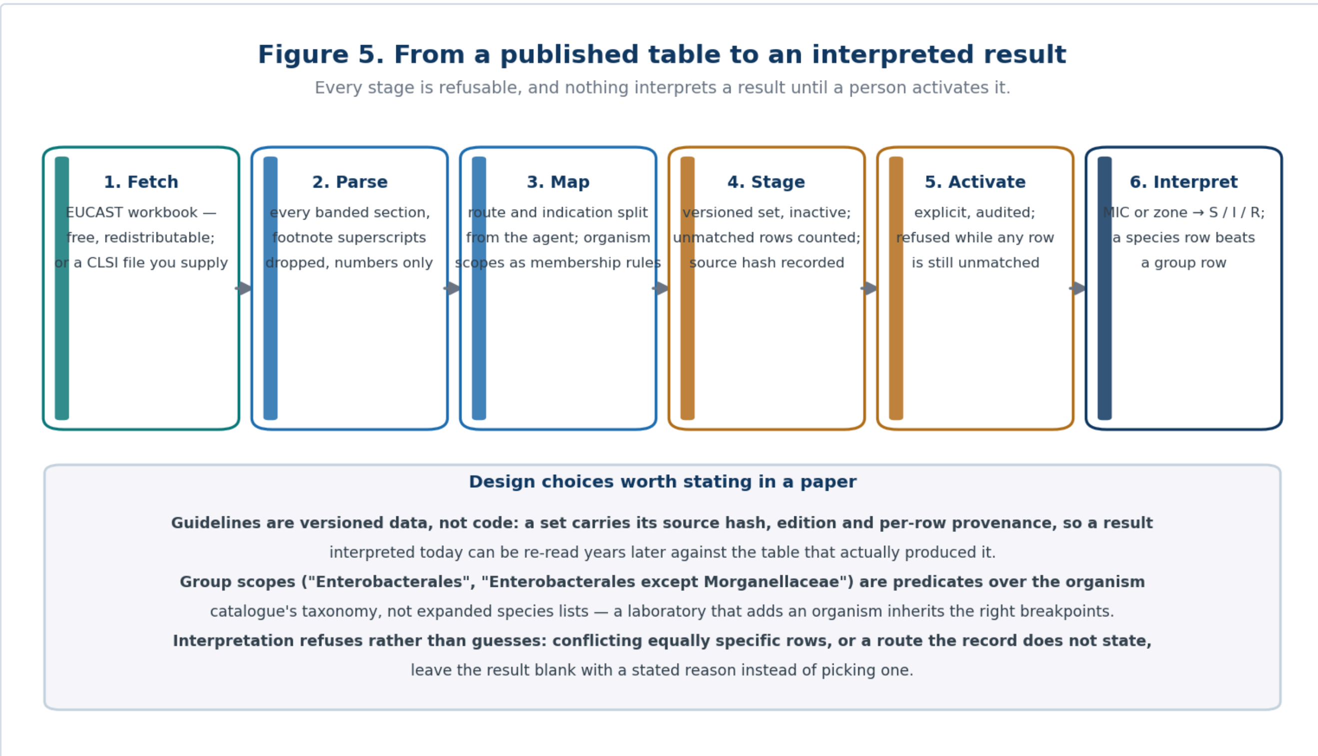

Figure 5 — from a published table to an interpreted result.

Its properties:

- Guidelines are versioned data. A staged set carries its source hash, edition, row provenance and import report, so a result interpreted in 2026 can be re-read in 2030 against the table that actually produced it.

- **Organism scopes are membership rules**, not expanded species lists. EUCAST writes "Enterobacterales", "Coagulase-negative staphylococci", "Enterobacterales except Morganellaceae"; those are stored as predicates over the organism catalogue's taxonomy. A species-specific row always beats a group row for the same organism.
- **Refusal over guessing.** Conflicting equally specific rows, a route the record does not state, or a missing matching row all produce a blank result with a stated reason.
- **Activation is a human act.** A set with unmatched rows cannot be activated at all.

Alternatives considered. Compiling breakpoints into the application makes every guideline revision a software release, which no laboratory can wait for. Accepting a spreadsheet without provenance makes a result unauditable. The staging-plus-activation design is the middle path: data-driven, but never live until somebody takes responsibility for it.

24. Deduplication

Two modes, chosen per analysis and always shown with the result:

- `firstPatientOrganism` — **first isolate per patient per organism** within the window. This is the CLSI M39 / WHO GLASS convention.
- `allIsolates` — every isolate, for laboratory workload and quality questions.

Why first-isolate is the default. Repeat isolates from the same patient are not independent observations: a single long ICU admission with daily cultures can contribute dozens of resistant isolates and shift a national rate on its own. Reporting resistance without deduplication systematically overestimates it in exactly the settings where the sickest patients are — which is where policy attention lands.

Why the alternative is kept. A laboratory asking "how much work did we do?" or "how complete is our AST?" needs every isolate. The mode is therefore a parameter of the question, not a global setting.

25. Indicators

The priority indicators follow WHO GLASS pathogen–antimicrobial combinations: MRSA; vancomycin-non-susceptible enterococci; third-generation-cephalosporin-non-susceptible *E. coli* and *K. pneumoniae*; carbapenem-non-susceptible Enterobacterales, *P. aeruginosa* and *Acinetobacter* spp.; fluoroquinolone-resistant *E. coli*; penicillin-non-susceptible *S. pneumoniae*.

Each is a numerator and a denominator, both reported. Non-susceptible (R + I) is used where GLASS uses it, and the tested-with agent set is stated, because "carbapenem resistance" computed over meropenem alone and over meropenem-or-imipenem-or-ertapenem are different numbers and only one of them matches the neighbouring country's.

26. Outbreak detection

Method: Kulldorff's space–time permutation scan statistic, case-only, prospective by default.

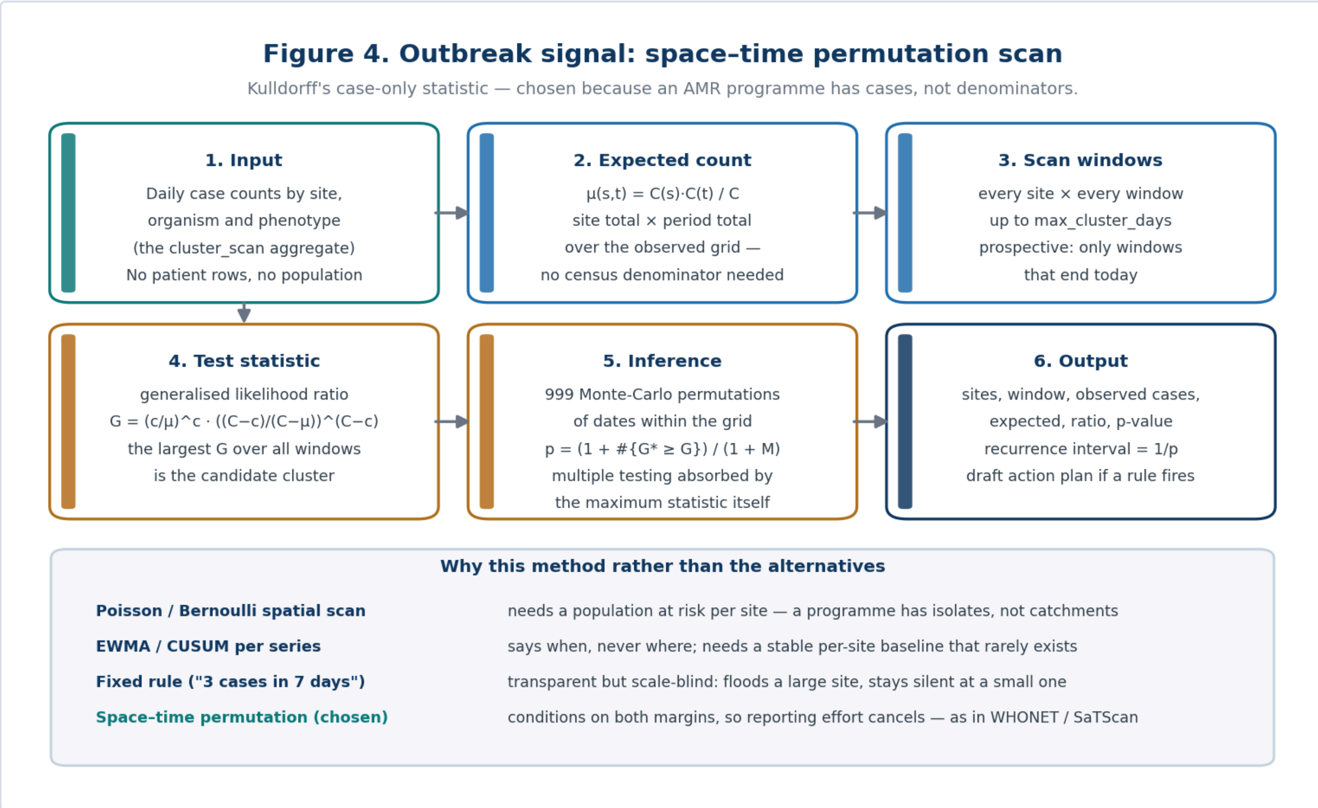

Figure 4 — the scan pipeline and the alternatives it was chosen over.

For each candidate cluster — a site (or set of sites) and a time window up to `max_cluster_days` — the expected count is the product of that location's total and that period's total divided by the grand total. The test statistic is the generalised likelihood ratio, and significance comes from Monte-Carlo permutation of dates within the observed grid (999 by default), so the multiple-testing burden of scanning thousands of windows is handled by the distribution of the *maximum* statistic rather than by a correction applied afterwards. Output includes the recurrence interval (1/p), which is what an operational programme can act on: "a cluster this extreme would be seen by chance about once every N days".

Why case-only. The Poisson and Bernoulli scan statistics need a population at risk per location. A national AMR programme does not have one: it has isolates from laboratories with different catchments, different testing rates and different referral patterns. Conditioning on both margins removes exactly that — a site that reports twice as much of everything contributes no signal, which is the property that makes the method usable across heterogeneous reporting.

Why not the alternatives.

Method	Why not
Fixed rule ("3 cases in 7 days")	Transparent but scale-blind: floods large sites, silent at small ones
EWMA / CUSUM per series	Detects *when* but not *where*; needs a stable per-site baseline
Poisson / Bernoulli spatial scan	Needs a denominator population AMRIT does not have
Unsupervised clustering	No inferential statement; a cluster is always found

Limitations, stated. The scan detects excess relative to the rest of the observed data, so a uniform national rise is invisible to it — that is what the trend metrics are for. It is sensitive to `baseline_days` and `max_cluster_days`, so both travel with every signal. And it operates on aggregate counts, so it can point a programme at a place and a period, never at a patient.

27. Privacy engineering

Four mechanisms, layered:

1. **Aggregate-only federation.** The wire contract carries counts and rates. A middleware guard rejects any response payload carrying patient identifiers, so a bug in a laboratory build cannot become a national disclosure.
2. **k-anonymity floor.** Cells below `AMRIT_K_ANONYMITY_FLOOR` (default 5) are suppressed before storage, not before display — a suppressed number is never written down.
3. **Coarsening.** Patient residence is stored to a coarsened postal geography; facility coordinates are site-level and consent-gated. Patient coordinates do not exist.
4. **Retention and erasure.** Row-level data expires on the profile's schedule; a subject's rows can be erased on request, and the erasure is itself auditable.

Why not de-identified record sharing? Because de-identified AMR records are re-identifiable in practice: a rare organism, a date and a district are frequently unique. Keeping rows in the laboratory and shipping only answers removes the question rather than mitigating it.

28. Federation protocol design

The laboratory **pulls** questions (`GET /v1/poll` , long-poll, with a WebSocket nudge for immediacy) and posts answers (`POST /v1/respond`). Both credentials — bearer token and out-of-band site token — are required on every request.

Why pull rather than push. A pushing laboratory must hold a queue, a retry policy, a credential for the server and an outbound firewall exception; when it is offline the programme learns nothing about why. A pulling laboratory needs only outbound HTTPS, answers only questions it has allow-listed, and its absence is visible as a missing answer rather than as silence.

Why not a central warehouse. A warehouse is the design that makes every other privacy control a promise. The moment rows are centralised, the protection is policy; while they stay local, it is architecture.

Why the two-token scheme. One token that both authenticates and travels the same channel as enrolment can be replayed by whoever intercepts enrolment. Splitting the factors across two channels means a compromise of either one alone is not enough.

29. Dashboard computation

Metrics are declared in a catalogue (`metrics/catalog.py`) as key, query type, filters and presentation. A refresh dispatches the de-duplicated set of wire queries the catalogue implies, waits for the batch, then computes each metric from *that batch only* and writes a `KPISnapshot` per metric and scope. Snapshots carry their computation time, source and site count, so a dashboard tile can always answer "as of when, from how many sites".

Scopes are canonical (`country`, `admin:<level>`, `site`) and derived from the country profile's tier chain, so a one-tier and a five-tier country use the same code path.

30. Limitations and threats to validity

Stated plainly, because a paper needs them:

- **Denominator heterogeneity.** Resistance rates depend on which isolates a laboratory tests and refers. Coverage metrics (sites reporting, AST completeness) are published alongside the rates for that reason.
- **Breakpoint version effects.** A guideline revision can move a national rate without any change in biology. Every result carries the set that produced it.
- **Deduplication sensitivity.** First-isolate versus all-isolate changes rates materially; comparisons across systems must match modes.
- **Suppression bias.** k-anonymity suppression removes small cells, which are not missing at random — sparse regions lose more.
- **Signal ≠ outbreak.** A scan statistic identifies statistical excess; confirmation is an epidemiological act.

Appendix A — Commands

```
# Desktop
pnpm --dir app install && pnpm --dir app dev      # development
pnpm --dir app run check && pnpm --dir app test   # types, lint, tests

# Portal
python3 manage.py migrate
python3 manage.py bootstrap                       # first administrator
python3 manage.py seed_demo [--country IND] [--strict]
python3 manage.py refresh_snapshots --live
python3 manage.py purge_sites --apply             # clears the registry
python3 manage.py purge_dashboard_data --apply    # clears snapshots and results

# Documentation
python3 tools/build_manual_figures.py
AMRIT_PORTAL_URL=… AMRIT_PORTAL_USER=… AMRIT_PORTAL_PASSWORD=… \
  app/node_modules/.bin/electron tools/capture_portal.mjs
python3 tools/build_manual.py
```

Appendix B — Environment

Variable	Meaning
`AMRIT_COUNTRY_PROFILE`	Country profile id (`IN` , `TESTLAND` , …). Must be passed to the container
`AMRIT_SEED_DEMO`	`1` seeds the demonstration pack — published passwords, demo only
`AMRIT_K_ANONYMITY_FLOOR`	Minimum cell size before suppression (default 5)
`AMRIT_QUERY_TTL_SECONDS`	How long a dispatched query stays answerable
`AMRIT_LONGPOLL_MAX_WAIT` , `AMRIT_LONGPOLL_TICK`	Long-poll hold time and poll granularity
`AMRIT_REFRESH_WAIT_SECONDS`	How long a live refresh waits for answers
`AMRIT_ENROLMENT_PICKUP_TTL_SECONDS`	Lifetime of a one-time pickup token (default 24 h)

Appendix C — Roles and dashboards

Role	Dashboard	Scope
Super administrator, programme administrator	Country	National
Policy maker, public health expert	Country	National
Epidemiologist, researcher	Epidemiology	National, with signals
Administrative health officer	Administrative	The viewer's own unit
Hospital administrator	Hospital	One facility
Press, citizen	Public summary	National, suppressed

Appendix D — Metric catalogue (extract)

`res_*` resistance rates per pathogen-agent pair · `phen_cre`, `phen_mdr` phenotypes · `burden_isolates`, `burden_organism_mix`, `burden_specimen_mix` · `antibiogram` · `cov_sites_reporting`, `cov_sites_online`, `cov_geo`, `cov_ast_completeness`.

Appendix E — Demonstration credentials

Seeded only by `seed_demo`, with passwords published in this repository. Never enable on a deployment holding real data.

User	Role
`superadmin`	Super administrator
`policy_maker`	Policy maker
`epidemiologist`	Epidemiologist
`researcher`	Researcher
`public_health`	Public health expert
`state_officer`, `district_officer`	Administrative health officer
`hospital_admin`	Hospital administrator
`press`, `citizen`	Public views